

Contextual Information Seeking for Active Inference.

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1 Introduction

Humans face the challenge of survival in complex environments with limited cognitive resources. Although they are capable of simulating complex future scenarios to act purposefully, in many cases evolution instead has shown that behaving habitually also leads to its survival while not allocating as many cognitive resources compared to complex planning. A recent approach from the computational psychiatries has shown how an individual might achieve the balancing between these two modes of behavior. The Balancing-Contextual-Control (BCC) model from ... introduces an abstract context state, which the individual first has to infer to decide whether it should perform planning, which is computationally demanding, or if it suffices to act habitually. An example for this mechanism could be a person who commutes from its home to its workplace regularly. After a short time of doing so, she would not need think about how to get there anymore. Taking the right turns will become habitual and the person is then cognitively free to think about other things than the exact route to her workplace. A context change would occur when there is construction work on the route that blocks the road. In this case, she should realize this context change and actively plan another route. Now, the person cannot rely anymore on computationally cheap habitual behavior but must take an effort to perform planning. In other words, she has inferred that the context has changed and switches her mode from habitual behavior to planning. The BCC model has shown to explain data from ... using this mechanism.

In line with this argument of surviving while saving cognitive resources, it might be plausible that the individual does not solely rely on the environment occasionally giving hints about the context but it rather actively tries to find out about the current context by itself. Relating to the commuting example, the person might have decided to listen to the local radio—not only because she likes the 90s popsongs playing there but also because of the current traffic news given there. In other words, she is curious about whether she can save resources and just take the same route like every day or if there is a reason (e.g., construction work) that requires her to think about another route.

This curiosity was formally described by a recent theory from the computational neurosciences which is Active Inference. It is based on the premise that an individual has a model of the environment that allows it to form probabilistic beliefs about it. Being curious about something in the environment then means to resolve uncertainty about the respective beliefs by acting purposefully to seek information.

In this thesis I will do a step forward at uniting the two principles (1. context dependent balancing between habitual behavior and planning and 2. curiosity) that both together could allow the individual to save cognitive resources in even more effective ways. Specifically, the resulting behavioral model will be curious about the context it is currently in and acts purposefully to resolve uncertainty about this context. Importantly,

this attempt will not completely unite the two principles outlined above but will only focus on the mechanism of finding out about the context as a consequence of curiosity. A next step, which is not part of this thesis, could then add habitual and planning modes that depend on the context. In other words the resulting model might describe why the person in the commuting example switches on the radio to learn about possible constructions works on its route but not what exactly she is then supposed to do with this new information, i.e., the discovered context does not yet allow for the balancing of habitual behavior and planning in this work. Moreover, the object to be developed in this thesis can be viewed from two perspectives. First, it might be seen as a hypothesis about behavior resulting from brain function that can be evaluated using real empirical data regarding its plausibility. Second, the resulting model can be viewed as an (artificial) agent that interacts with an environment and thus directly allows for investigating the model's behavior. In this work, it is central to prove that the model shows the expected behavior, which is why the developed object will be viewed from the perspective of an agent.

In the following I will outline how this model will be developed in this thesis. As mentioned before, context dependent balancing of habitual behavior and planning was already worked out in the BBC model and curiosity is an integral part of the theory of Active Inference. Consequently, the approach in this work will be based upon these two lines of research. So far, I have not detailed what Active Inference is about. In short, it is about the kind of behavior that is expected from a living creature that faces the challenge of survival in an a chaotic environment. These behaviors then allow for the development of artificial Active Inference agents, that are assumed to possess the same basic principles (that apparently includes curiosity) that ensure the survival of real biological individuals. Conveniently, the BCC model itself is strongly inspired by Active Inference but misses out on the curiosity part. Following this, the model to be developed here can be viewed from the following two perspectives (not to be confused with the two views of understanding it either as a hypothesis of behavior and as an artificial agent): First, it can be seen as the BCC model that gets expanded by the curiosity mechanism for its context but leaves out on the balancing between habitual behavior and planning. Second, it might also be understood as an Active Inference agent (that naturally comes with curiosity), that models curiosity over an added context in its model. In this work, I regularly refer to this second perspective. This view requires us to understand in detail what Active Inference is about, which is explained in the following.

Active Inference

Active Inference is an emerging cognitive theory about behavior and brain function of many living creatures. It assumes that individuals must solve the problem of survival in chaotic environments that potentially desintegrate their physical bodies and thus end their existences. Furthermore, the theory takes the perspective that these dangerous environmental effects on the individual's body are dependent on the current state of the environment itself, e.g., getting burn wounds from touching a hot plate is dependent on

the state of that plate itself (is it hot or cold?), which is part of the environment. Unfortunately, the individual does not immediately know about the state of the environment. As a remedy for this problem it has developed sensors (e.g., eyes, ears, nociceptors etc.) to make observations that yield information about the current environmental state. In other words, the theory makes the central assumption, that in general the state of the environment is hidden but has an effect on the individual in two ways: First, it may influence the individual's body in harmful ways (e.g., changing the chemical composition of skin as a result of burn wounds), and second by generating observations that the animal can sense (e.g., nociceptor-based pain as a result of heat) that are somehow related to the true but unobservable state of the environment. Intuitively, the individual should then regularly receive observations that are related to these unobservable environmental states that do not have dangerous effects on its body. As a counter example, consider an individual, that regularly receives observations that are related to environmental states that also usually have the effect of ending the individual's existence. Such a creature would just not exist.

So far, we have bluntly stated that animals have sensors that allow them to receive information about unobservable environmental states. This knowledge is not immediately useful for the animal—except in the case when it has the means to exert influence over these states. Just as we considered the existence of sensors as given, we can also notice that animals can perform actions upon the environment (e.g., turning the hot plate off). Influencing the environment's hidden states via actions means the individual implicitly controls the kind of observations it receives in return. Because it needs observations that hint at environmental states that are not dangerous for its continued existence it must somehow infer these hidden states based on observations to be able to influence them in purposeful ways using actions. In conclusion, first, the individual must possess a model that relates observations to environmental hidden states to be able to infer the latter given the current observation. Second, it then should also have a model of the relation between its actions and how they influence hidden states in the environment over time.

Regarding the first requirement, this is usually formalized as a generative model $p(s, o)$ that formulates the probabilistic dependencies between hidden states s and observations o . Because observations are influenced by the environment—i.e., they are initially out of control of the individual—it maintains a stable model by explicitly modeling the prior $p(s)$ and also how it relates to observations via the likelihood $p(o|s)$, i.e., the generative model is available to the individual explicitly as these two terms. Upon receiving a specific observation $\hat{o}$ the individual can then infer the likely hidden states that led to this observation by computing (approximating) $p(s|\hat{o})$. In a second step this result can then be used to infer useful actions (this will be explained later in this section).

First, we must consider how the animal could compute $p(s|\hat{o})$ as a basis for future actions. In principle this can be achieved by leveraging Bayes theorem via $p(s|\hat{o}) = \frac{p(s)p(o|s)}{\int p(s)p(o|s) ds}$. In reality the integration in the denominator usually is not tractable for the animal because it would need to consider a possibly arbitrary amount of hidden states s . As a remedy, the individual can do something different: Instead of exactly calculating

$p(s|\hat{o})$ it can optimize a variational distribution $q(s)$ that assumes the true $p(s|\hat{o})$ after optimization has finished. Note, that the loss function that drives this optimization already takes into account $\hat{o}$ such that it does not need to be made an explicit dependency in $q(s)$, i.e., $q(s)$ is implicitly dependent on $\hat{o}$. In the following, I will introduce this loss function that enables the animal to optimize $q(s)$ step by step.

In principle, we can use a distance measure to quantify the (dis)similarity between the true posterior $p(s|\hat{o})$ and the variational distribution $q(s)$. In Active Inference this is achieved by the Kullback-Leibler (KL) divergence

$$D_{\text{KL}}[q(s) \parallel p(s|\hat{o})] = \int q(s) \ln \frac{q(s)}{p(s|\hat{o})} ds \quad (1.1)$$

that becomes 0 exactly when $q(s)$ assumes the true $p(s|\hat{o})$. Conversely, the quantity becomes larger the more dissimilar both distributions are. In theory, this quantity (KL divergence between $q(s)$ and $p(s|\hat{o})$) suffices as loss function for optimizing $q(s)$. In reality, to goal of finding the distribution $p(s|\hat{o})$ via $q(s)$ is the reason why this optimization is done in the first place, i.e., $p(s|\hat{o})$ is not available and thus Eq. (1.1) is not computable.

There is a solution for this problem. Instead of minimizing (by optimizing $q(s)$) the KL divergence with respect to the true but unavailable posterior $p(s|\hat{o})$, it can be minimized with respect to the whole generative model $p(s, o) = p(s)p(o|s)$, which apparently has the same effect. This circumstance will be explained in the following. First, we consider the KL divergence between $q(s)$ and the generative model:

$$D_{\text{KL}}[q(s) \parallel p(s)p(\hat{o}|s)] = \int q(s) \ln \frac{q(s)}{p(s)p(\hat{o}|s)} ds. \quad (1.2)$$

Eq. (1.2) clearly shows that the individual is able to compute this loss function because it has access to $p(s)$ and $p(\hat{o}|s)$. Next, Eq. (1.2) will be rearranged to reveal the kind of objective that is measured by it:

$$\begin{aligned} D_{\text{KL}}[q(s) \parallel p(s)p(\hat{o}|s)] &= \int q(s) \ln \frac{q(s)}{p(s)p(\hat{o}|s)} ds \\ &= \int q(s) \ln \frac{q(s)}{p(s, \hat{o})} ds \\ &= \int q(s) \ln \frac{q(s)}{p(s|\hat{o})p(o)} ds \\ &= \int q(s) \ln \frac{q(s)}{p(s|\hat{o})} ds - \ln p(\hat{o}) \end{aligned} \quad (1.3)$$

The last line in Eq. (1.3) makes it obvious that the (computable) loss function of Eq. (1.2) consists of the uncomputable loss function of Eq. (1.1) which is appended by the summand $-\ln p(\hat{o})$. In other words, by minimizing Eq. (1.2), the individual achieves to optimize $q(s)$ with respect to $p(s|\hat{o})$ but the loss function will not become 0 when both distributions are equal but instead settle at the value given by the constant $-\ln p(\hat{o})$. Summarizing, the animal is able to compute the true posterior $p(s|\hat{o})$ (via optimizing

$q(s)$) by minimizing a quantity that yields $-\ln p(\hat{o})$ at its minimum. This residue is known as "surprise" of the observation $\hat{o}$ and the quantity itself is called Variational Free Energy (VFE) and given the letter F

$$F[q(s), \hat{o}] = \int q(s) \ln \frac{q(s)}{p(s)p(\hat{o}|s)} ds = \int q(s) \ln \frac{q(s)}{p(s|\hat{o})} ds - \ln p(\hat{o}), \quad (1.4)$$

that is a functional of the variational distribution $q(s)$ and the current observation $\hat{o}$. Therefore F is also referred to as an upper bound on surprise, i.e, the bound is reached from above exactly when $q(s)$ assumes the true posterior $p(s|\hat{o})$.

In conclusion, the individual should minimize this quantity F to infer the hidden environmental states, which then allows it to act in a way that support its existence.

2 blüb

Note, that by defining the (indispensable) generative model, we have created an intricate situation. This is because now the individual explicitly assigns probabilities to hidden environmental states as a consequence of the necessity of knowing their probabilistic relationship to observations, i.e., the prior $p(s)$ exists. This alone is not interesting. What makes this situation special is the animal's dependency on very specific hidden states that do not pose a threat to its continued existence. Following this, the marginal $p(s)$ should then describe exactly these states, i.e., it should assign high probabilities to hidden states that are not dangerous. In other words $p(s)$ might be viewed as the ecological niche of the animal in the environment that it should not leave if it "wants" to survive. As a side note, do not understand $p(s)$ as an exact image of that environmental niche, rather it might be viewed as a functional abstraction or homomorphism of the real environmental conditions.

Understanding $p(s)$ as the animal's niche in the environment has intriguing consequences. In short, the individual must receive specific observations o that are related to specific states s (via $p(s, o)$), that have a high probability as given by $p(s)$, i.e., the observations must hint that the animal did not leave its niche. This is then formally given as the marginal (evidence) $p(o) = \int p(s, o) ds$ of its generative model that describes how much it "wants" to see certain observations as given by their respective probabilities, i.e., favourable observations have a higher probability in $p(o)$.

As a result the animal should then somehow act in certain ways that result in observations as defined by $p(o)$. Crucially, acting purposeful requires the animal to quantify this affordance. In principle this quantity is already given as the probability $p(o)$. In reality, the individual only has access to $p(s)$ and $p(o|s)$. While these terms allow for the computation of $p(o)$ via $p(o) = \int p(s)p(o|s) ds$, in practice this integration is oftentimes intractable because of the possibly arbitrary amount of states s . As a remedy for this problem, the individual can instead compute a quantity called Variational Free Energy (VFE) that can serve as a proxy to measure how well an observation fits $p(o)$.

3 blub

The thesis aims at developing an Active Inference Agent that has a context state as part of its generative model and acts goal-oriented to resolve uncertainty about this context. This development task is achieved as a synthesis of three approaches: The first is Active Inference which allows for the development of behaving Agents that are intrinsically driven to be curious about the world and act purposeful to resolve uncertainty about hidden states in their generative models. The second approach entails the recently introduced Bayesian-Contextual-Control model that is strongly inspired by Active Inference and includes an additional context state in its generative model that allows for dynamic balancing between habitual and goal-directed behavior, although this balancing is beyond this work. The third is a certain modeling approach of such agents that represents their generative models as Forney factor graphs and views the behavior-underpinning inference process as custom messages sent in this graph. The three approaches are detailed in the following.

Active Inference

Active Inference is an emerging theory in the neurosciences that accounts for cognitive function and behavior of most living creatures. It is based on the premise that an individual survives by maintaining its body's integrity. This active maintenance is necessary because it is situated in an environment that could potentially destroy this integrity and thus end the individual's existence. In the case of animals that are capable of higher level reasoning this might be achieved as follows. Animals have developed sensors (e.g., eyes, ears, etc.) that allow them to acquire information (observations) about the outside world. Additionally, evolution has equipped them with preferences (or aversions) for certain observations. These preferences have developed in parallel with its physical bodies and are optimized to secure its integrity. As an example, consider the fear of heights. The animal does not exactly know what falling off a cliff means to the structural integrity of its organelles (e.g., tissue, bones, etc.) – the only point of reference it immediately has is strong fear. This evolutionary optimized heuristic is what is meant with preferences or aversions (e.g., the sight of height) for certain observations. Conclusively, the animal somehow "wants" to ensure to only receive the preferred observations. Luckily, most individuals have developed means to do exactly that, which is acting upon the environment. By executing actions and thus influencing the environment the animal can effectively control what it observes. Following this line of argument, it then may choose its actions wisely to receive exactly these observations that it likes (or avoid the ones it dislikes) according to its evolutionary optimized preferences. Doing this wisely requires the individual to "understand" which actions lead to

which observations, i.e., it must have a model of the chain of causality starting from its actuators (e.g., limbs) going through the environment and ending at its sensors (e.g., eyes). This understanding comes in the form of a generative model. Let us view this situation in more detail. The individual is equipped with a model of the causality chain that is tuned (by evolution) to predict preferred observations at its "end". Specifically, this chain does not allow all paths of causal steps that might be possible in the outside environment but only the ones that lead to preferred observations that underwrite the animals existence. Formally, the individual's (probabilistic) generative model is given by the likelihood $p(o|\Omega)$ and prior $p(\Omega)$, where o models observations as the end of the causality chain and Ω makes up for the rest of that chain including tractable paths through the environment starting from its actors that lead to preferred observations o . The maxime of the animal then becomes to receive observations o that are most likely under its generative model. Crucially, this requires it to "know" how likely certain observations actually are which necessitates the computation of the marginal evidence $p(o)$. Unfortunately, this is usually not possible in finite time because it would need to compute $\int p(o|\Omega)p(\Omega) d\Omega$, which requires integration over a possibly intractable amount of states Ω . Fortunately, there is a remedy for this problem. Instead of exactly calculating the probability of observations $p(o)$, it can calculate a proxy quantity that enables it to approximately evaluate how likely a specific observation under its generative model is. This quantity is the variational free energy (VFE). To discover this energy in more detail, we must first understand that the intractable probability $p(o)$ can also be given as a surprise quantity $-\ln p(o)$, that assigns low and high surprise values to likely and unlikely observations respectively. The variational free energy then becomes an upper bound on surprise, i.e., it is a quantity that needs to be minimized by the individual and reaches the exact surprise $-\ln p(o)$ at its minimum. The VFE is given by

$$F[o, q(\Omega)] = \int q(\Omega) \ln \frac{q(\Omega)}{p(o|\Omega)p(\Omega)} d\Omega. \quad (3.1)$$

Next, the different parts of Eq. (3.1) will be investigated. The variational free energy F is a functional (not a function), that has two arguments o and q . o is the observation, for which F assigns a value, that evaluates the observation's fitting under the generative model, i.e., it is the upper bound on surprise $-\ln p(o)$. The second argument of F is the variational distribution $q(\Omega)$, that is subject to optimization. Basically, the variational free energy functional F is a loss function, that when minimized with respect to the parameters of $q(\Omega)$ assumes the true surprise $-\ln p(o)$. In this optimal case, $q(\Omega)$ also becomes equal to the true posterior $p(\Omega|o)$. This becomes evident when rearranging Eq. (3.1) as follows:

$$F[o, q(\Omega)] = \int q(\Omega) \ln \frac{q(\Omega)}{p(\Omega|o)} d\Omega - \ln p(o). \quad (3.2)$$

Here, F becomes minimal exactly when $\int q(\Omega) \ln \frac{q(\Omega)}{p(\Omega|o)} d\Omega$ assumes 0, which is the case when $q(\Omega) = p(\Omega|o)$. Then, only the term $-\ln p(o)$ remains as a non-zero contribution to F , i.e., F is then equal to the true surprise $-\ln p(o)$ and can thus be used by the

individual to evaluate how likely a certain observation is under its generative model. Now, there are two aspects to consider here. First, the individual must minimize F to be able to evaluate $-\ln p(o)$. Second, the individual also "wants" observations, that have a low surprise $-\ln p(o)$, i.e., a high probability under its generative model. Elegantly, this second requirement is also taken into account by F because it will become even smaller when the surprise also becomes smaller. These insights lead to the central conclusion, that the individual should minimize its variational free energy F as the premise that underwrites its existence.

There is something more intricate about what is actually achieved by the minimization of F . It can be useful to view this process as physically embodied. Following this, the generative model itself might be realized by the wiring of the animal's nervous system, i.e., most importantly the brain. Building on this, optimizing the variational distribution $q(\Omega)$ with respect to the variational free energy might be seen as the inference process of the animal as a result of which it forms beliefs about the outside world and also how to act in useful ways. These (stabilized) beliefs are then modeled by $q(\Omega)$ after it has been optimized. Concluding, minimizing F can be equated to the process of thinking itself.

Next, we investigate how action and planning might be realized.

4 Methods

4.1 Inference via Message Passing

In this section we introduce the general approach how to derive a message passing algorithm from a probabilistic model that enables inference for the agents presented later. Specifically, we start with a probabilistic model and show that it has a representation as a Forney factor graph. Then we introduce belief distributions that exist in the same graph. Next, we formulate a variational (Bethe) free energy functional over the probabilistic model and the belief distributions. Building on this, we identify constraints on the belief distributions. These constraints then expand the free energy functional to form a Lagrangian. This will be the starting point for deriving the messages. Specifically, by finding local stationary solutions of the Lagrangian we identify these local terms as messages on the factor graph. Furthermore, these messages can directly be used to update the belief distributions. The updated distributions are the result of the message passing procedure. Finally, the order in which messages are sent will be addressed leading to a specific schedule of execution. The following is not to be understood as a rigorous formulation nor does it cover all edge cases. Rather it should give an intuition of the procedure using a simple example. The main task in this work of deriving functional agents will be presented in later sections.

4.1.1 The Probabilistic Generative Model

The probabilistic generative model represents a system (e.g. the brain or a model of behavior) using a range of variables, including their dependencies with probability distributions to account for uncertainties within the system. To exemplify the principal method of building behaving agents later in this work, we will now resort to the simple probabilistic model $p(x, y)$ which can be factorized into a prior $p(x)$ and a likelihood $p(y|x)$.

4.1.2 The Forney Factor Graph Representation

A Forney factor graph (FFG) is a neat tool to represent a factorized probabilistic model where probabilistic factors (e.g., $p(x)$) become nodes and variables (e.g., x) become edges. Our example model $p(x, y)$ has two variables x, y and is factorized into two factors $p(x)$ and $p(y|x)$. This leads to an FFG that equally has two edges and two nodes as shown in Fig. 4.1.

In this work we utilize a version of an FFG that requires every edge to be terminated by a node on both ends, resulting in a (terminated) Forney factor graph. To this end

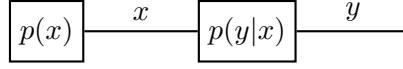


Figure 4.1: FFG of the model $p(x, y)$.

our model can be extended by a factor 1 resulting in $p(x, y) = p(x)p(y|x) \cdot 1$ with the (terminated) FFG shown in Fig. 4.2

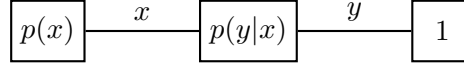


Figure 4.2: FFG of the model $p(x, y)$ with terminated edges.

To facilitate future derivations we rename node factors with functions f that have indices as unique identifiers. Following this shift, the three nodes become $p(x) = f_a(x)$, $p(y|x) = f_b(x, y)$, and $1 = f_c(y)$ resulting in the FFG in Fig. 4.3. Note that f_c has an argument y because it is connected to the edge y but always results into 1.

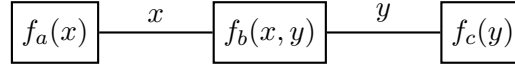


Figure 4.3: FFG of the model $p(x, y)$ with renamed factors $(f_a(x), f_b(x, y), f_c(y))$.

4.1.3 Belief Distributions

The central task of behaving agents presented later in this work will be inference. Inference basically is the process of calculating posterior distributions of the probabilistic model. For example when variable y is set to be an observation variable and gets fixed to a specific observation $\hat{y}$, inference corresponds with calculating the posterior $p(x|\hat{y})$. In principle this (Bayesian inversion) can exactly be done using Bayes Theorem with $p(x|\hat{y}) = \frac{p(x)p(\hat{y}|x)}{p(\hat{y})}$. In this work we instead resort to an approximate method that relies on variational (approximate) distributions named q . Referring to the little example, we are not exactly calculating $p(x|\hat{y})$ but instead $q(x)$ will be subject to optimization as an approximation of $p(x|\hat{y})$. In general, q distributions are driven by any change in the whole (possibly very large) FFG (e.g. fixing multiple variables, other constraints, etc.). In other words these approximate distributions might be seen as beliefs that react to new evidence anywhere in the graph. Following this the terms approximate posteriors/distributions and beliefs (distributions) will be used synonymously in this work. In the Forney factor graph beliefs exist for all edges and nodes resulting in the FFG in Fig. 4.4. Here node beliefs are indexed with the respective indices of the factors (a, b, c) and edge beliefs get indices i, j . Edge beliefs have a clear interpretation in terms of approximate posteriors of the respective variables. In contrast, node beliefs might be seen as necessary to propagate information through nodes – although they form posterior beliefs (possibly over multiple variables) similar to edge beliefs. In some cases node

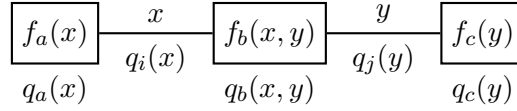


Figure 4.4: FFG of the model $p(x, y)$ with belief distributions.

beliefs can be interpreted as an approximation to joint beliefs over variables, while edge beliefs are interpreted as marginal posteriors over variables. Having y as an observations renders its corresponding variational distribution a delta distribution that enforces that the variables gets "clamped" to a specific observation $\hat{y}$, i.e., $q_j(y) = \delta_{y, \hat{y}}$. Note, that we use the Kronecker delta because distributions will be discrete in this work. The resulting FFG is shown in Fig. 4.5.

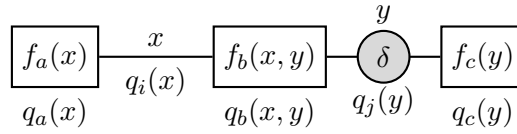


Figure 4.5: FFG of the model $p(x, y)$ with an observation as a delta constraint.

4.1.4 The (Bethe) Free Energy Functional

The belief distributions q introduced above are optimized with respect to a loss function which is the variational free energy functional F (to be minimized) (also known as the negative evidence lower bound (ELBO)) of the FFG. Fundamentally, the variational free energy is the (Kullback-Leibler) divergence between the joint belief distribution $q(x, y)$ and the probabilistic model $p(x, y)$ with

$$F = D_{\text{KL}}(q(x, y) \parallel p(x, y)) = \sum_{x, y} q(x, y) \log \frac{q(x, y)}{p(x, y)}, \quad (4.1)$$

where $q(x, y) = \frac{q_a(x)q_b(x, y)q_c(y)}{q_i(x)q_j(y)}$ and $p(x, y) = f_a(x)f_b(x, y)f_c(y)$.

This functional can be further decomposed into local free energy terms from each node as well as local entropy terms that correspond to the edges. The free energy functional of the FFG in Fig. 4.4 then becomes

$$\begin{aligned} F = & \underbrace{\sum_x q_a(x) \log \frac{q_a(x)}{f_a(x)} + \sum_{x, y} q_b(x, y) \log \frac{q_b(x, y)}{f_b(x, y)} + \sum_y q_c(y) \log \frac{q_c(y)}{f_c(y)}}_{\text{Local Free Energy Terms}} \\ & + \underbrace{\sum_x q_i(x) \log \frac{1}{q_i(x)} + \sum_y q_j(y) \log \frac{1}{q_j(y)}}_{\text{Correcting Entropy Terms}}. \end{aligned} \quad (4.2)$$

Note that we assumed discrete distributions (e.g., categorical), leading to sums instead of integrals. Each local free energy at a node measures the Kullback-Leibler divergence between the local belief distribution q and the local factor f . Importantly, these local free energies contain entropy terms that correspond to the edges entropy it is connected to. As an example consider the local free energy term $\sum_x q_a(x) \log \frac{q_a(x)}{f_a(x)}$ which can be decomposed into $-(\sum_x q_a(x) \log q_a(x)) - \sum_x q_a(x) \log f_a(x)$. The term $-(\sum_x q_a(x) \log q_a(x))$ is the entropy of edge i , i.e., it is equivalent to $-(\sum_x q_i(x) \log q_i(x))$. Similarly node b also contains this entropy of edge i as part of $-(\sum_{x,y} q_b(x,y) \log q_b(x,y))$. Consequently, having two nodes for every edge results into overcounting of edge entropies, i.e., it is subtracted twice. To correct for this one entropy term is added for each edge in F , e.g., $\sum_x q_i(x) \log \frac{1}{q_i(x)}$. Moreover, we still used $q_j(y)$ in the equation. In following derivations this is substituted by the concrete distribution $\delta_{y,\hat{y}}$. The herein used free energy functional is also known as the Bethe free energy.

4.1.5 The Lagrangian of the Free Energy

In this section we will find out that the belief distributions q will not solely be found with respect to the variational free energy F , but instead they will be determined by minimizing a Lagrangian L . Previously we defined a Forney factor graph that contains variables, factors, and belief distributions for all of which we could define a unified free energy. For the following task of deriving a message passing algorithm we first need to focus on the constraints on beliefs distributions implied by the FFG. The central aspect here is that, first all belief distributions must meet marginalization constraints with respect to neighboring belief distributions with

$$q_i(x) \stackrel{!}{=} q_a(x) \quad (4.3) \quad q_j(y) \stackrel{!}{=} \sum_x q_b(x,y) \quad (4.5)$$

$$q_i(x) \stackrel{!}{=} \sum_y q_b(x,y) \quad (4.4) \quad q_j(y) \stackrel{!}{=} q_c(y). \quad (4.6)$$

Equivalently, they all must be normalized to be valid probability distributions with

$$\sum_x q_a(x) \stackrel{!}{=} 1 \quad (4.7) \quad \sum_x q_i(x) \stackrel{!}{=} 1 \quad (4.10)$$

$$\sum_{x,y} q_b(x,y) \stackrel{!}{=} 1 \quad (4.8) \quad \sum_y q_j(y) \stackrel{!}{=} 1. \quad (4.11)$$

$$\sum_y q_c(x) \stackrel{!}{=} 1 \quad (4.9)$$

Adding these constraints to F leads to the Lagrangian

$$\begin{aligned}
L = F &+ \sum_x \lambda_{ia}(x)(q_i(x) - q_a(x)) + \sum_x \lambda_{ib}(x)(q_i(x) - \sum_y q_b(x, y)) \\
&+ \sum_y \lambda_{jb}(y)(q_j(y) - \sum_x q_b(x, y)) - \sum_y \lambda_{jc}(y)(q_j(y) - q_c(y)) \\
&+ \psi_a(\sum_x q_a(x) - 1) + \psi_b(\sum_{x,y} q_b(x, y) - 1) + \psi_c(\sum_y q_c(y) - 1) \\
&+ \psi_i(\sum_x q_i(x) - 1) + \psi_j(\sum_y q_j(y) - 1),
\end{aligned} \tag{4.12}$$

where λ and ψ are Lagrange multipliers that enforce marginalization and normalization respectively. Crucially, instead of finding the belief distributions q by minimizing the variational free energy F , they will be determined by minimizing the Lagrangian L .

4.1.6 Messages and Belief Updates

For the message passing algorithm we now derive the messages necessary for updating the belief distributions from the Lagrangian L by finding local stationary solutions for the belief distributions. To do this, we use the functional (or variational) derivative, as L is a functional and our quantities of interest are distributions, not variables. The derivation requires multiple steps. First, L is partially differentiated with respect to each variational distribution q , then variational derivatives are set to 0, and solved for q^* :

$$\frac{\delta L}{\delta q_a(x)} = \log q_a(x) + 1 - \log f_a(x) - \lambda_{ia}(x) + \psi_a \stackrel{!}{=} 0 \tag{4.13}$$

$$\Rightarrow q_a^*(x) = f_a(x) \exp(\lambda_{ia}(x)) \exp(-\psi_a - 1). \tag{4.14}$$

By a similar argument it follows that

$$\Rightarrow q_b^*(x, y) = f_b(x, y) \exp(\lambda_{ib}(x)) \exp(\lambda_{jb}(y)) \exp(-\psi_b - 1) \tag{4.15}$$

$$\Rightarrow q_c^*(y) = f_c(y) \exp(\lambda_{jc}(y)) \exp(-\psi_c - 1) \tag{4.16}$$

$$\Rightarrow q_i^*(x) = \exp(\lambda_{ia}(x)) \exp(\lambda_{ib}(x)) \exp(\psi_i - 1) \tag{4.17}$$

$$\Rightarrow q_j^*(y) = \exp(\lambda_{jb}(y)) \exp(\lambda_{jc}(y)) \exp(\psi_j - 1). \tag{4.18}$$

Second, we partially solve the system of equations (Eqs. (4.14) to (4.18)) with respect to the normalization constraints with

$$\sum_x q_a^*(x) \stackrel{!}{=} 1 \quad (4.19)$$

$$\sum_x f_a(x) \exp(\lambda_{ia}(x)) \exp(-\psi_a - 1) = 1 \quad (4.20)$$

$$\exp(-\psi_a - 1) \sum_x f_a(x) \exp(\lambda_{ia}(x)) = 1 \quad (4.21)$$

$$\exp(-\psi_a - 1) = \frac{1}{\sum_x f_a(x) \exp(\lambda_{ia}(x))} \quad (4.22)$$

$$\Rightarrow q_a^*(x) = f_a(x) \exp(\lambda_{ia}(x)) \frac{1}{\sum_x f_a(x) \exp(\lambda_{ia}(x))} \quad (4.23)$$

$$\Rightarrow q_a^*(x) \propto f_a(x) \exp(\lambda_{ia}(x)). \quad (4.24)$$

The term $\exp(-\psi_a - 1)$ effectively becomes the normalization constant for $q_a^*(x)$. Instead of doing this derivation repeatedly in subsequent sections, I will directly use the proportional sign $\propto$ when defining belief distributions. Analogously we obtain

$$\Rightarrow q_b^*(x, y) \propto f_b(x, y) \exp(\lambda_{ib}(x)) \exp(\lambda_{jb}(y)) \quad (4.25)$$

$$\Rightarrow q_c^*(y) \propto f_c(y) \exp(\lambda_{jc}(y)) \quad (4.26)$$

$$\Rightarrow q_i^*(x) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{ib}(x)) \quad (4.27)$$

$$\Rightarrow q_j^*(y) \propto \exp(\lambda_{jb}(y)) \exp(\lambda_{jc}(y)). \quad (4.28)$$

Third, we solve the remaining system of equations (Eqs. (4.24) to (4.28)) with respect to the marginalization constraints with

$$q_a^*(x) \stackrel{!}{=} q_i(x) \quad (4.29)$$

$$f_a(x) \exp(\lambda_{ia}(x)) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{ib}(x)) \quad (4.30)$$

$$\Rightarrow \exp(\lambda_{ib}(x)) \propto f_a(x). \quad (4.31)$$

Similarly, it can be shown that

$$\Rightarrow \exp(\lambda_{ia}(x)) \propto \sum_y f_b(x, y) \exp(\lambda_{jb}(y)) \quad (4.32)$$

$$\Rightarrow \exp(\lambda_{jc}(y)) \propto \delta_{y, \hat{y}} \quad (4.33)$$

$$\Rightarrow \exp(\lambda_{jb}(y)) \propto \delta_{y, \hat{y}}. \quad (4.34)$$

Note, that we substituted the Kronecker delta $\delta_{y, \hat{y}}$ for all $q_j(y)$ in Eqs. (4.31) to (4.34). Finally we substitute the factors f with the real distributions in Eqs. (4.31) to (4.34)

leading to

$$\exp(\lambda_{ib}(x)) \propto p(x) \quad (4.35)$$

$$\exp(\lambda_{ia}(x)) \propto \sum_y p(x|y) \exp(\lambda_{jb}(y)) \quad (4.36)$$

$$\exp(\lambda_{jc}(y)) \propto \delta_{y,\hat{y}} \quad (4.37)$$

$$\exp(\lambda_{jb}(y)) \propto \delta_{y,\hat{y}}. \quad (4.38)$$

By following the steps above we have derived the messages (Eqs. (4.35) to (4.38)) and also how the belief distributions (Eqs. (4.24) to (4.28)) are formed by them.

4.1.7 Messages in the Factor Graph

Having derived the four messages above, we can now view them inside the factor graph. Their names give a clear interpretation of their position and direction of information flow in the graph. For example, the message $\exp(\lambda_{ia}(x))$ has indices i and a , meaning that its location in the FFG is on the edge i and points towards node a . The resulting graph with all messages is given in Fig. 4.6.

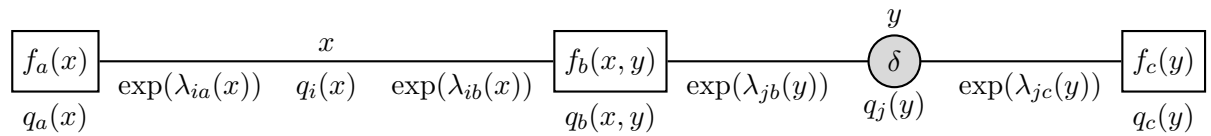


Figure 4.6: FFG of the model $p(x, y)$ with messages.

Using the full message names renders the FFG visually overloaded. Instead we use abbreviations ($\overleftarrow{\mu}$, $\overrightarrow{\mu}$) with arrows clearly indicating the direction of information flow with $\overleftarrow{\mu}(x) = \exp(\lambda_{ia}(x))$, $\overrightarrow{\mu}(x) = \exp(\lambda_{ib}(x))$, $\overleftarrow{\mu}(y) = \exp(\lambda_{jb}(y))$, and $\overrightarrow{\mu}(y) = \exp(\lambda_{jc}(y))$. The resulting FFG is given in Fig. 4.7.

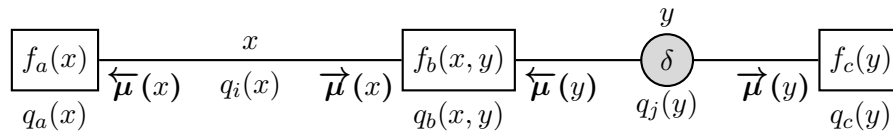


Figure 4.7: FFG of the model $p(x, y)$ with μ messages.

4.1.8 Schedule

At this point, we have determined the messages for the message passing in the FFG but have not shown how the resulting algorithm works exactly. Recall that its core purpose

is the calculation of the variables' (edge) belief distributions (Eqs. (4.27) and (4.28)) with

$$q_i(x) \propto \overleftarrow{\mu}(x) \overrightarrow{\mu}(x) \quad (4.39)$$

$$q_j(y) \propto \overleftarrow{\mu}(y) \overrightarrow{\mu}(y), \quad (4.40)$$

where they are the product of the two messages on the respective edges in the FFG. Crucially, next we will show that the four messages depend on each other. When substituting μ in Eqs. (4.35) to (4.38) we get

$$\overleftarrow{\mu}(x) \propto \sum_y p(y|x) \overleftarrow{\mu}(y) \quad (4.41)$$

$$\overrightarrow{\mu}(x) \propto p(x) \quad (4.42)$$

$$\overleftarrow{\mu}(y) \propto \delta_{y,\hat{y}} \quad (4.43)$$

$$\overrightarrow{\mu}(y) \propto \delta_{y,\hat{y}}. \quad (4.44)$$

This interdependence of messages ($\overleftarrow{\mu}(x)$ depends on $\overleftarrow{\mu}(y)$) leads to the following order of execution in Algorithm 1. Note, that $\overrightarrow{\mu}(y)$ and $\overleftarrow{\mu}(y)$ can be also be evaluated

Algorithm 1 Message Passing Algorithm of the Model $p(x, y)$

- 1: **evaluate** $\overrightarrow{\mu}(x)$, $\overrightarrow{\mu}(y)$ and $\overleftarrow{\mu}(y)$
 - 2: **evaluate** $\overleftarrow{\mu}(x)$
 - 3: **evaluate** $q_i(x)$ and $q_j(y)$
-

in step 2 and in general there can exist multiple valid schedules. Moreover, there is no order in each evaluation step, e.g., for $\overrightarrow{\mu}(x)$, $\overrightarrow{\mu}(y)$ and $\overleftarrow{\mu}(y)$ in step 1.

4.1.9 Remarks

Previously we mentioned that resulting belief distributions (e.g., $q_i(x)$ and $q_j(y)$) are approximations of their true posteriors. While in general this is the case, in our example the beliefs are in fact exact and therefore equal to these true posteriors. This relates to a certain choice of distributions q . More specifically, for nodes that connect at least two variables (only $f_b(x, y)$ in our example) we did not assume independence of participating variables in the respective q distribution (here $q_b(x, y) \neq q_b(x)q_b(y)$). This leads to a message passing algorithm that keeps all these local dependencies between variables intact and enables exact inference as long as the FFG is acyclic. In our example this situation specifically leads to the Bethe free energy. Moreover the resulting message passing algorithm is also known as the Sum-Product algorithm or Belief Propagation.

4.2 Agents

In this section I present the four agents that are developed in this work. First, I detail what agents entail. In general, agents are entities that are in constant exchange with an

external environment. They receive observations from and act upon the environment. In this work observations and actions occur at discrete timesteps and are signified via the variables o_t and a_t respectively at the current timestep t (Fig. 4.8). The environ-

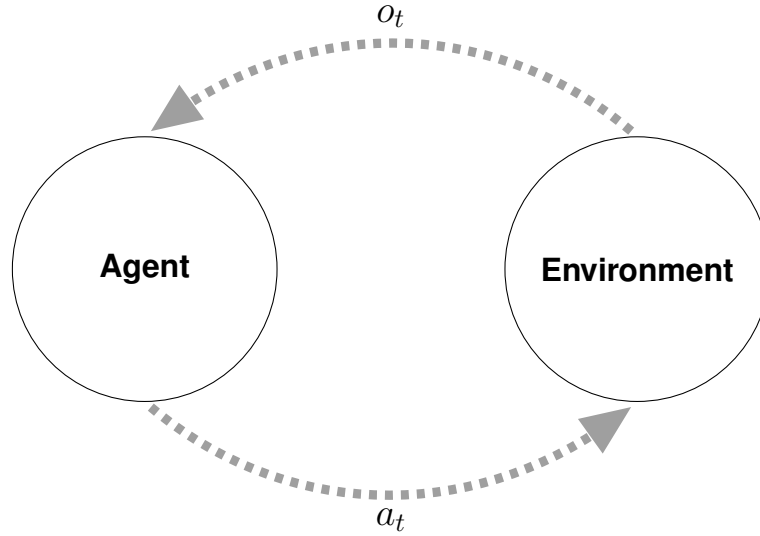


Figure 4.8: Perception-Action-Loop between the agent and the environment. The environment sends observations o_t to the agent. Vice versa, the agent sends actions a_t to the environment. The current timestep is t .

ment generates observations in possibly arbitrary ways but the agent is assumed to have certain mechanisms that process these observations and trigger actions. Specifically, agents are assumed to have a preference for certain observations that underwrite their ecological niche in the environment. Moreover, they are equipped with the means to understand (simulate) the chain of causality from hidden environmental causes to its sensors, i.e., they have of model how the state of the environment leads to certain observations. Even more, they also have a model of the causal chain of these unobservable (hidden) environmental states through time and how they can be influenced via actions. In total, this allows them to plan a sequence of actions that leads to preferred observations. Additionally, it allows them to keep past observations as a memory and enables them to retrospectively draw conclusions about past events in the light of new observations from the present.

This mechanism rests upon a generative model that probabilistically relates observations, actions, and the (discrete) succession of hidden states. Formally, it is given by a likelihood $p(o|s)$ that states how observations o are generated by hidden states s and by a transition model $p(s_{t+1}|s_t, a_t)$ that defines how hidden states s change from timestep t to $t + 1$ given an action a_t . Actions possess an action prior $p(a)$ and preferences for observations are encoded by the goal prior $\tilde{p}(o)$ that assigns higher probabilities to preferred observations and action

The four distributions (likelihood, transition model, action prior, and goal prior) are

building blocks for a temporal rollout (from timestep 0 to T), which formally leads to a Partially Observable Markov Decision Process (POMDP). In later simulations there will be three timesteps, i.e., $T = 2$, and hidden states at $t = 0$ are initially distributed with $p(s_0)$. The resulting POMDP is given by

$$\begin{aligned} p(o_0, o_1, o_2, s_0, s_1, s_2, a_0, a_1) = & \tilde{p}(o_0)p(o_0|s_0)p(s_0) \\ & \tilde{p}(o_1)p(o_1|s_1)p(s_1|s_0, a_0)p(a_0) \\ & \tilde{p}(o_2)p(o_2|s_2)p(s_2|s_1, a_1)p(a_1) \end{aligned} \quad (4.45)$$

and has a factor graph representation (Fig. 4.11). The POMDP in Eq. (4.47) and the

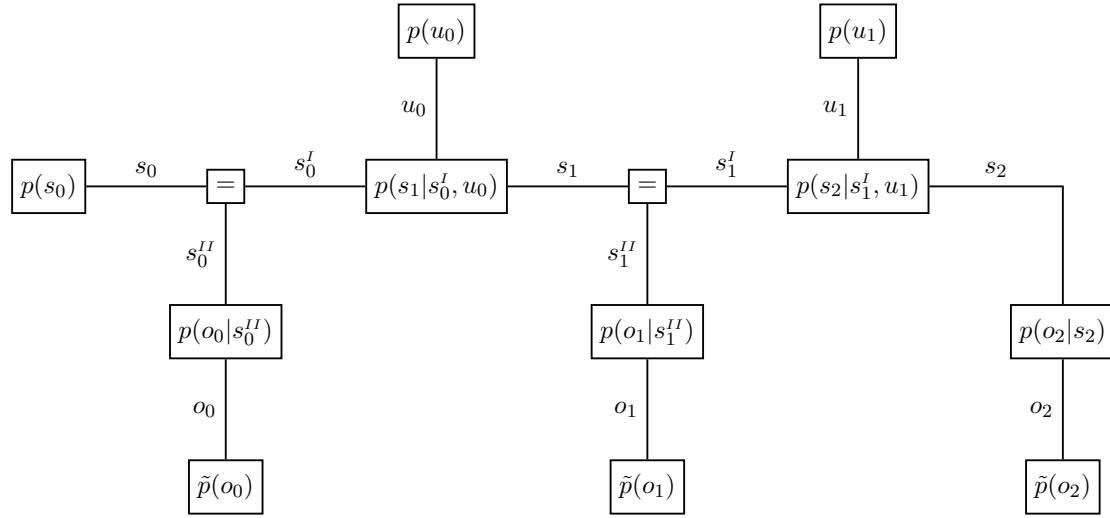


Figure 4.9: Factor Graph of the POMDP from the two non-Context Agents.

corresponding factor graph in Fig. 4.11 will be the basis for the BFE- and GFE agents that are introduced in following sections.

Next, I expand the POMDP of Eq. (4.47) to develop a basis for the Context-BFE- and Context-GFE agents. These agents do not only assume that observations are generated by hidden states s but additionally assume a hidden context state c that also influences observations. Moreover, the context state does not change for each timestep (like hidden states s) and thus is assumed to be persistent during multiple timesteps. In later simulations, there will only be a single context state during the succession of three regular hidden states s . The corresponding POMDP is given by

$$\begin{aligned} p(o_0, o_1, o_2, s_0, s_1, s_2, c, a_0, a_1) = & \tilde{p}(o_0)p(o_0|s_0, c)p(s_0)p(c) \\ & \tilde{p}(o_1)p(o_1|s_1, c)p(s_1|s_0, a_0)p(a_0) \\ & \tilde{p}(o_2)p(o_2|s_2, c)p(s_2|s_1, a_1)p(a_1) \end{aligned} \quad (4.46)$$

and entails a prior over the context $p(c)$ similarly to the initial prior over the hidden state $p(s_0)$. The corresponding factor graph of Eq. (4.46) is given in Fig. 4.10.

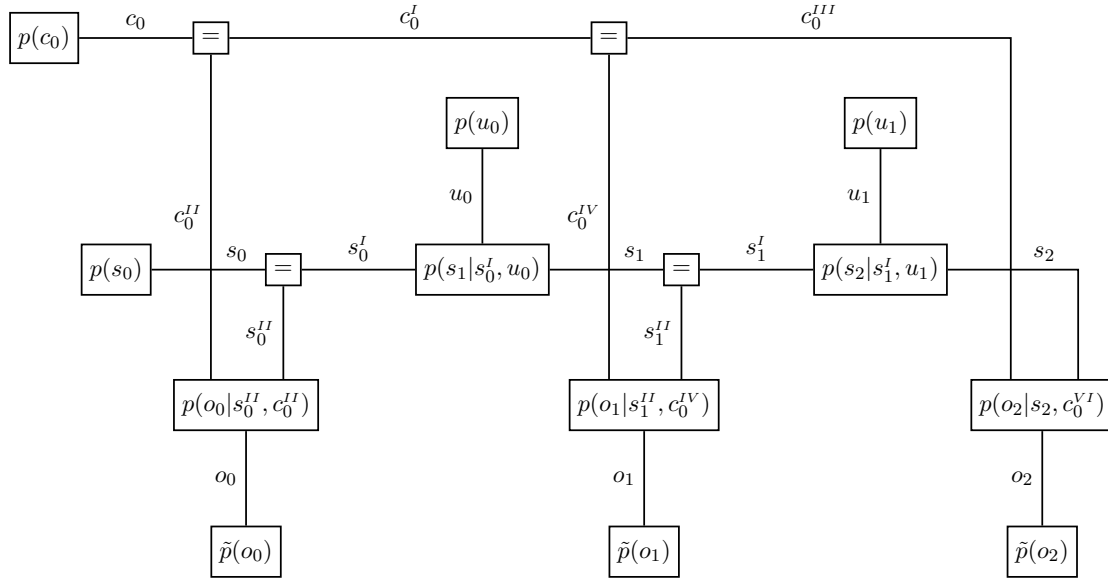


Figure 4.10: Factor Graph of the POMDP from the two Context Agents.

Note, that in both POMDPs (Eqs. (4.46) and (4.47)) some variables occur in more than two factors (e.g., s_0 in $p(s_0)$, $p(o_0|s_0)$ and $p(s_1|s_0, a_0)$ in Eq. (4.47)). When representing these POMDPs as factor graphs a special situation arises in which these variables (modeled by edges) should be connected to more than two factor nodes. Crucially, factor graphs are allowed to have exactly two nodes for each edge which requires an artificial duplication of variables to make them available for more than two factor nodes (e.g., s_0 is duplicated to s_0^I and s_0^{II}). Duplicated variables are forced to be equivalent via the special equality node (indicated by the = sign) they are all connected to. Importantly, this situation is only a formal requirement of factor graphs and has no consequences for the results of inference processes presented in following sections.

4.2.1 Bethe Free Energy (BFE) Agent

In this section I present the generative model of the BFE Agent and also derive a message passing algorithm on its factor graph representation based on the recipe from Section 4.1 that enables it with inference. Aside from this, it will also be explained how the agent can perceive and act which makes its behavioral repertoire complete. These insights will also be the basis for the three agents presented in later sections of this work.

The Temporal Generative Model

As outlined previously the temporal generative model is the substrate that enables the agent to understand the environment, integrate observations, keep memories, plan into

the future, select useful actions, and perform inference. Throughout this work the time horizon of temporal generative models will be $T = 3$. The resulting POMDP is then given by

$$p(o_0, o_1, o_2, s_0, s_1, s_2, a_0, a_1) = \tilde{p}(o_0)p(o_0|s_0)p(s_0)\tilde{p}(o_1)p(o_1|s_1)p(s_1|s_0, a_0)p(a_0) \quad (4.47)$$

$$\tilde{p}(o_2)p(o_2|s_2)p(s_2|s_1, a_1)p(a_1).$$

The temporal generative model of Eq. (4.47) has a clear representation as a Forney factor graph (Fig. 4.11). The distributions are modeled by nodes while variables exist

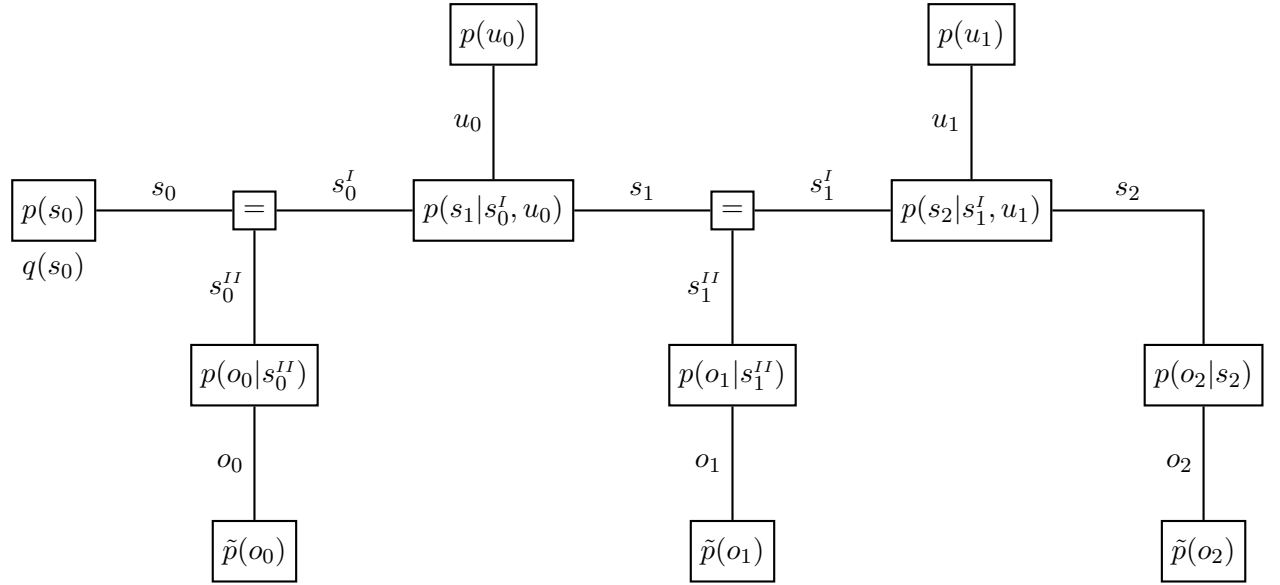


Figure 4.11: Factor graph of the temporal generative model.

on edges. Related to this, there is a specialty about factor graphs that needs to be addressed. Edges are allowed to have a maximum number of two of connected nodes. The variables s_0 and s_1 are used in more than two distributions each ($p(s_0)$, $p(s_1|s_0, u_0)$, $p(o_0|s_0)$) and ($p(s_1|s_0, u_0)$, $p(o_0|s_0)$, $p(s_2|s_1, u_1)$) respectively. This leads to a situation where the edges of both variables should be connected to three nodes each. To accommodate for this problem equality nodes (=) are introduced that exist at the respective T-shaped edges in the graph. Effectively, the single T-shaped edge is split into three distinct edges. This leads to a technical duplication of the respective variables in the graph ($s_0 \rightarrow s_0, s_0^I, s_0^{II}$) and ($s_1 \rightarrow s_1, s_1^I, s_1^{II}$). Importantly, the duplication of these variables has no effect on the inference process presented later.

Inference

Having the temporal generative model represented as a Forney factor graph allows for a straightforward derivation of the inference procedure based on messages that are sent in the graph. Recall, that the reason for doing inference are the belief distributions q

that exist for all edges and nodes (see q distributions Fig. 4.22). During inference these will converge to approximate (marginal) posteriors distributions that eventually enable the agent to sample useful actions from. In accordance with Section 4.1 inference requires a loss function to be minimized. This function is a Lagrangian L that entails a variational free energy functional and normalization and marginalization constraints. For the temporal generative model it is given by

. Here, the constraints are subsumed into the variable C . Next, local stationary points of the belief distributions can be derived and using the marginalization constraints we arrive at terms that can be interpreted as messages on the graph's edges that also allows for the final computation of the belief distributions. These derivations strictly follow Section 4.1. Instead of individually deriving all messages in the whole graph, I will show the derivations for two distinct cases that occur. First, for messages sent from equality nodes, and second for the remaining messages. Fundamentally they all follow the same principles and are named Bethe messages because they stem from a pure Bethe free energy. This clarification will become increasingly important in later sections of the other three agents where the underlying free energy will include additional constraints, thus it is not pure Bethe anymore, such that resulting messages will be totally different.

To show the kind of messages sent from equality nodes we will consider the left equality node in the graph that has an edge to node $p(s_0)$. The central aspect about equality nodes is that they have a factor function that guarantees that all connected variables (on edges) maintain equal values. Concretely, this is enforced by a product of Kronecker deltas (in the case of discrete distributions). Here the factor function of the left equality node is then given by

$$f_{=} = \delta_{x,x^I} \delta_{x,x^{II}}. \quad (4.48)$$

For the following derivation, only parts of the Lagrangian that relate to this node will be necessary. This partial Lagrangian is given by

. Following the recipe of Section 4.1 the

Terminating-Node Messages

This section introduces the messages that are sent by terminating nodes in the factor graph. The general approach is as follows. Consider the FFG in Fig. 4.12 where we are interested in $\vec{\mu}(x)$. First, we need to identify the free energy functional of the FFG,

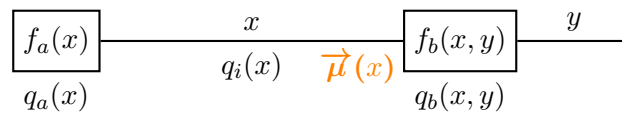


Figure 4.12: Example FFG with terminating-node message (orange).

which is given by

$$F = \sum_x q_a(x) \log \frac{q_a(x)}{f_a(x)} + \sum_{x,y} q_b(x,y) \log \frac{q_b(x,y)}{f_b(x,y)} + \sum_x q_i(x) \log \frac{1}{q_i(x)}. \quad (4.49)$$

Next, the belief distributions must meet the normalization and marginalization constraints

$$\sum_x q_a(x) \stackrel{!}{=} 1 \quad (4.50) \quad q_i(x) \stackrel{!}{=} q_a(x) \quad (4.53)$$

$$\sum_{x,y} q_b(x,y) \stackrel{!}{=} 1 \quad (4.51) \quad q_i(x) \stackrel{!}{=} \sum_y q_b(x,y), \quad (4.54)$$

$$\sum_x q_i(x) \stackrel{!}{=} 1 \quad (4.52)$$

which leads to the Lagrangian

$$\begin{aligned} L = & \sum_x q_a(x) \log \frac{q_a(x)}{f_a(x)} + \sum_{x,y} q_b(x,y) \log \frac{q_b(x,y)}{f_b(x,y)} + \sum_x q_i(x) \log \frac{1}{q_i(x)} \\ & + \psi_a \left(\sum_x q_a(x) - 1 \right) + \psi_b \left(\sum_{x,y} q_b(x,y) - 1 \right) + \psi_i \left(\sum_x q_i(x) - 1 \right) \\ & + \sum_x \lambda_{ia}(x) (q_i(x) - q_a(x)) + \sum_x \lambda_{ib}(x) (q_i(x) - \sum_y q_b(x,y)). \end{aligned} \quad (4.55)$$

Following this, we can now find local stationary solutions with respect to the three belief distributions $q_a(x)$, $q_b(x,y)$, and $q_i(x)$ with

$$\frac{\delta L}{\delta q_a(x)} = \log q_a(x) + 1 - \log f_a(x) - \lambda_{ia}(x) + \psi_a \stackrel{!}{=} 0 \quad (4.56)$$

$$\Rightarrow q_a^*(x) \propto f_a(x) \exp(\lambda_{ia}(x)) \quad (4.57)$$

$$\frac{\delta L}{\delta q_b(x,y)} = \log q_b(x,y) + 1 - \log f_b(x,y) - \lambda_{ib}(x) + \psi_b \stackrel{!}{=} 0 \quad (4.58)$$

$$\Rightarrow q_b^*(x) \propto f_b(x) \exp(\lambda_{ib}(x)) \quad (4.59)$$

$$\frac{\delta L}{\delta q_i(x)} = -\log q_i(x) - 1 + \lambda_{ia}(x) + \lambda_{ib}(x) + \psi_i \stackrel{!}{=} 0 \quad (4.60)$$

$$\Rightarrow q_i^*(x) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{ib}(x)) \quad (4.61)$$

and solve the system of equations (Eqs. (4.57), (4.59) and (4.61)) for $\vec{\mu}(x) (= \exp(\lambda_{ib}(x)))$ using the marginalization constraint $q_a^*(x) \stackrel{!}{=} q_i(x)$ with

$$q_a^*(x) \stackrel{!}{=} q_i(x) \quad (4.62)$$

$$f_a(x) \exp(\lambda_{ia}(x)) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{ib}(x)) \quad (4.63)$$

$$\Rightarrow \exp(\lambda_{ib}(x)) \propto f_a(x) \quad (4.64)$$

$$\Rightarrow \vec{\mu}(x) \propto f_a(x). \quad (4.65)$$

This derivation shows that the outgoing message from a terminating node to its neighbor node is the factor function of the terminating node, i.e., $f_a(x)$ in this example.

Following this the outgoing messages from the terminating nodes in the factor graph of the temporal generative model (Fig. 4.13) can now be clearly stated with

$$\vec{\mu}(s_0) \propto p(s_0) \quad (4.66)$$

$$\downarrow\mu(u_0) \propto p(u_0) \quad (4.67)$$

$$\downarrow\mu(u_1) \propto p(u_1) \quad (4.68)$$

$$\uparrow\mu(o_0) \propto \tilde{p}(o_0) \quad (4.69)$$

$$\uparrow\mu(o_1) \propto \tilde{p}(o_1) \quad (4.70)$$

$$\uparrow\mu(o_2) \propto \tilde{p}(o_2). \quad (4.71)$$

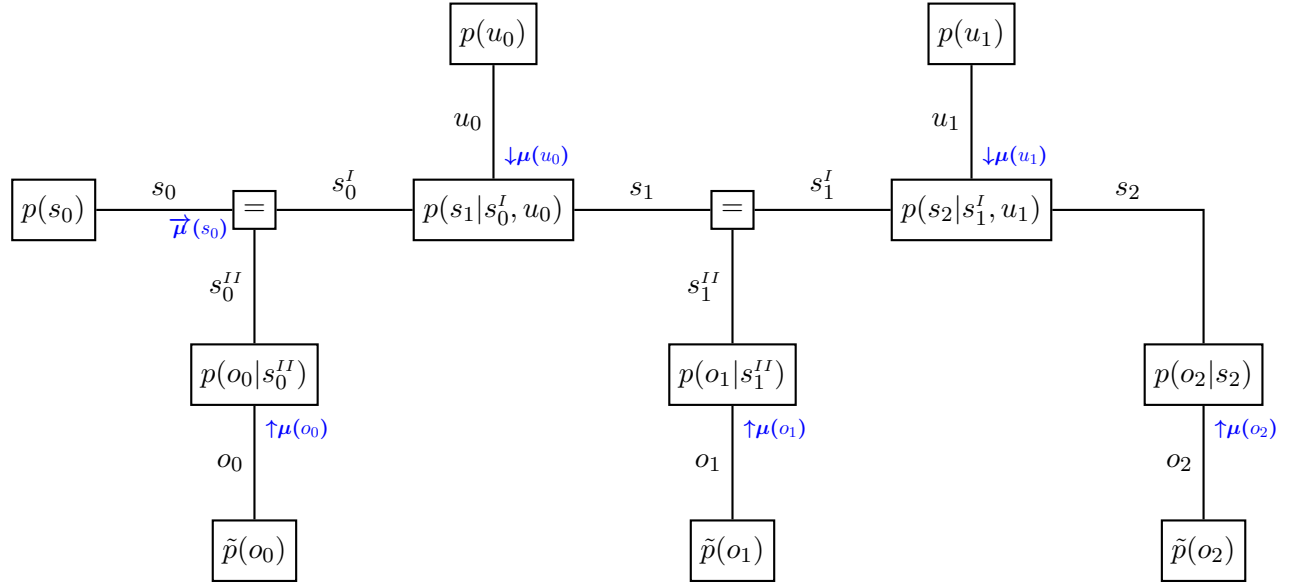


Figure 4.13: Factor graph of the temporal generative model with outgoing messages from terminating nodes (blue).

Bethe Messages

This section focuses on the messages sent by nodes whose local free energy is the Bethe free energy which enables exact local inference. In general we assume that the inference process in the factor graph is driven by a variational free energy, which means the resulting belief distributions are only approximations rather than the result of exact Bayesian inference. The reason for this situation is at the nodes themselves. More precisely, the resulting belief distributions will only be exact if all nodes in the graph propagate information, i.e., incoming messages without a loss. Such a loss of information can occur if a node locally assumes that its variables are stochastically independent. For example, for a node n with variables x and y this is true if $q_n(x, y) = q_n(x)q_n(y)$. Note that $f_n(x, y)$ may still model dependencies of x and y . In this situation the (mean-field) factorization $q_n(x)q_n(y)$ cannot model these dependencies and can therefore only be an approximation of the true local function $f_n(x, y)$. On the other hand,

if a node locally models these dependencies between its variables with its q distribution then information gets propagated without a loss and the node-local inference process becomes exact. In this case, the node locally assumes the Bethe free energy. Consider the factor graph in Fig. 4.14 where node b locally assumes the Bethe free energy F_b with

$$F_b = \sum_{x,y,z} q_b(x,y,z) \log \frac{q_b(x,y,z)}{f_b(x,y,z)}, \quad (4.72)$$

i.e., $q_b(x,y,z)$ respects the dependencies between x , y , and z . Now we determine the

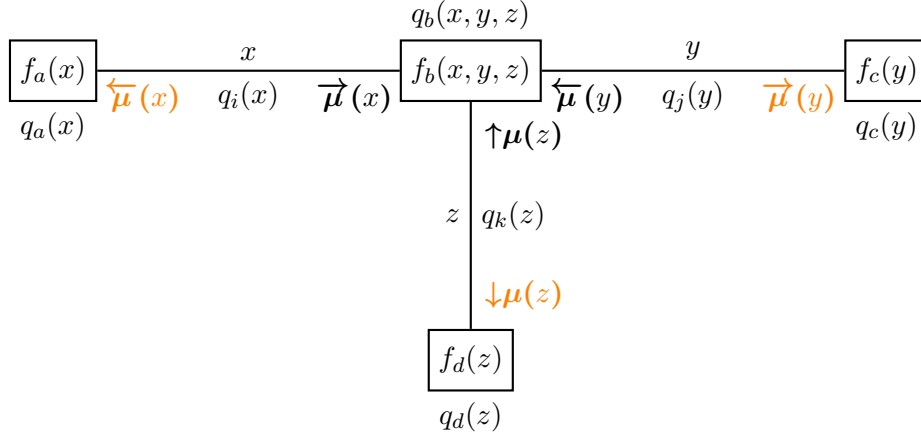


Figure 4.14: Example FFG with Bethe messages (orange).

messages $\overleftarrow{\mu}(x)$, $\overrightarrow{\mu}(y)$, and $\downarrow\mu(z)$ which are going out of node b to its neighbor nodes a , c , and d . First we must define the free energy functional of the FFG as given by

$$\begin{aligned} F = & \sum_{x,y,z} q_b(x,y,z) \log \frac{q_b(x,y,z)}{f_b(x,y,z)} + \sum_x q_a(x) \log \frac{q_a(x)}{f_a(x)} + \sum_y q_c(y) \log \frac{q_c(y)}{f_c(y)} \\ & + \sum_z q_d(z) \log \frac{q_d(z)}{f_d(z)} + \sum_x q_i(x) \log \frac{1}{q_i(x)} + \sum_y q_j(y) \log \frac{1}{q_j(y)} \\ & + \sum_z q_k(z) \log \frac{1}{q_k(z)}. \end{aligned} \quad (4.73)$$

Adding marginalization and normalization constraints to F leads to the Lagrangian

$$\begin{aligned}
L = F &+ \sum_x \lambda_{ib}(x)(q_i(x) - \sum_{y,z} q_b(x,y,z)) + \sum_y \lambda_{jb}(y)(q_j(y) - \sum_{x,z} q_b(x,y,z)) \\
&+ \sum_z \lambda_{kb}(z)(q_k(z) - \sum_{x,y} q_b(x,y,z)) + \sum_x \lambda_{ia}(x)(q_i(x) - q_a(x)) \\
&+ \sum_y \lambda_{jc}(y)(q_j(y) - q_c(y)) + \sum_z \lambda_{kd}(z)(q_k(z) - q_d(z)) + \psi_b(\sum_{x,y,z} q_b(x,y,z) - 1) \\
&+ \psi_a(\sum_x q_a(x) - 1) + \psi_c(\sum_y q_c(y) - 1) + \psi_d(\sum_z q_d(z) - 1) \\
&+ \psi_i(\sum_x q_i(x) - 1) + \psi_j(\sum_y q_j(y) - 1) + \psi_k(\sum_z q_k(z) - 1).
\end{aligned} \tag{4.74}$$

Solving the belief distributions in L for the stationary points then yields

$$\frac{\delta L}{\delta q_b(x,y,z)} = \log q_b(x,y,z) + 1 - \log f_b(x,y,z) - \lambda_{ib}(x) - \lambda_{jb}(y) - \lambda_{kb}(z) + \psi_b \stackrel{!}{=} 0 \tag{4.75}$$

$$\Rightarrow q_b^*(x,y,z) \propto f_b(x,y,z) \exp(\lambda_{ib}(x)) \exp(\lambda_{jb}(y)) \exp(\lambda_{kb}(z)). \tag{4.76}$$

Similarly

$$\Rightarrow q_a^*(x) \propto f_a(x) \exp(\lambda_{ia}(x)) \tag{4.77}$$

$$\Rightarrow q_c^*(y) \propto f_c(y) \exp(\lambda_{jc}(y)) \tag{4.78}$$

$$\Rightarrow q_d^*(z) \propto f_d(z) \exp(\lambda_{kd}(z)) \tag{4.79}$$

$$\Rightarrow q_i^*(x) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{ib}(x)) \tag{4.80}$$

$$\Rightarrow q_j^*(y) \propto \exp(\lambda_{jb}(y)) \exp(\lambda_{jc}(y)) \tag{4.81}$$

$$\Rightarrow q_k^*(z) \propto \exp(\lambda_{kb}(z)) \exp(\lambda_{kd}(z)). \tag{4.82}$$

Next, we use the marginalization constraint $\sum_{y,z} q_b^*(x,y,z) \stackrel{!}{=} q_i^*(x)$ to solve for the message sent by node b to node a (via edge i) with

$$\sum_{y,z} q_b^*(x,y,z) \stackrel{!}{=} q_i^*(x) \tag{4.83}$$

$$\sum_{y,z} f_b(x,y,z) \exp(\lambda_{ib}(x)) \exp(\lambda_{jb}(y)) \exp(\lambda_{kb}(z)) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{ib}(x)) \tag{4.84}$$

$$\Rightarrow \exp(\lambda_{ia}(x)) \propto \sum_{y,z} f_b(x,y,z) \exp(\lambda_{jb}(y)) \exp(\lambda_{kb}(z)) \tag{4.85}$$

$$\Rightarrow \overleftarrow{\mu}(x) \propto \sum_{y,z} f_b(x,y,z) \overleftarrow{\mu}(y) \uparrow \mu(z). \tag{4.86}$$

Analogously using the constraints $\sum_{x,z} q_b^*(x, y, z) \stackrel{!}{=} q_j^*(y)$ and $\sum_{x,y} q_b^*(x, y, z) \stackrel{!}{=} q_k^*(z)$ we obtain

$$\Rightarrow \vec{\mu}(y) \propto \sum_{x,z} f_b(x, y, z) \vec{\mu}(x) \uparrow\mu(z) \quad (4.87)$$

$$\Rightarrow \downarrow\mu(z) \propto \sum_{x,y} f_b(x, y, z) \vec{\mu}(x) \overleftarrow{\mu}(y). \quad (4.88)$$

This derivation reveals that outgoing messages from a node (b) that locally assumes the Bethe free energy is the factor function itself (f_b) multiplied with all incoming messages (e.g., $\vec{\mu}(x)$, $\uparrow\mu(z)$ in Eq. (4.87)) from its neighbors (except from the neighbor that is the addressee of the outgoing message) where all variables of the incoming messages get summed over (e.g., x and z in Eq. (4.87)).

Using this blueprint we can now clearly state the Bethe messages in the temporal generative model (Fig. 4.15) with

$$\overleftarrow{\mu}(s_0^I) = \sum_{u_0, s_1} p(s_1 | s_0^I, u_0) \downarrow\mu(u_0) \overleftarrow{\mu}(s_1) \quad \uparrow\mu(s_0^{II}) = \sum_{o_0} p(o_0 | s_0^{II}) \uparrow\mu(o_0) \quad (4.95)$$

$$(4.89) \quad \downarrow\mu(o_0) = \sum_{s_0^{II}} p(o_0 | s_0^{II}) \downarrow\mu(s_0^{II}) \quad (4.96)$$

$$\uparrow\mu(u_0) = \sum_{s_0^I, s_1} p(s_1 | s_0^I, u_0) \vec{\mu}(s_0^I) \overleftarrow{\mu}(s_1) \quad \uparrow\mu(s_1^{II}) = \sum_{o_1} p(o_1 | s_1^{II}) \uparrow\mu(o_1) \quad (4.97)$$

$$(4.90)$$

$$\vec{\mu}(s_1) = \sum_{s_0^I, u_0} p(s_1 | s_0^I, u_0) \vec{\mu}(s_0^I) \downarrow\mu(u_0) \quad \downarrow\mu(o_1) = \sum_{s_1^{II}} p(o_1 | s_1^{II}) \downarrow\mu(s_1^{II}) \quad (4.98)$$

$$(4.91) \quad \overleftarrow{\mu}(s_1^I) = \sum_{o_2} p(o_2 | s_2) \uparrow\mu(o_2) \quad (4.99)$$

$$\overleftarrow{\mu}(s_1^I) = \sum_{u_1, s_2} p(s_2 | s_1^I, u_1) \downarrow\mu(u_1) \overleftarrow{\mu}(s_2) \quad \downarrow\mu(o_2) = \sum_{s_2} p(o_2 | s_2) \downarrow\mu(s_2) \quad (4.100)$$

$$(4.92)$$

$$\uparrow\mu(u_1) = \sum_{s_1^I, s_2} p(s_2 | s_1^I, u_1) \vec{\mu}(s_1^I) \overleftarrow{\mu}(s_2) \quad (4.93)$$

$$\downarrow\mu(s_2) = \sum_{s_1^I, u_1} p(s_2 | s_1^I, u_1) \vec{\mu}(s_1^I) \downarrow\mu(u_1) \quad (4.94)$$

Note that we need the messages sent by equality nodes (gray) for the definitions of the Bethe messages. They will be properly explained in the next section of this work.

Equality-Node Messages

In this section I present the messages sent by equality nodes. As mentioned before edges in the factor graph are allowed to have a maximum number of connected nodes

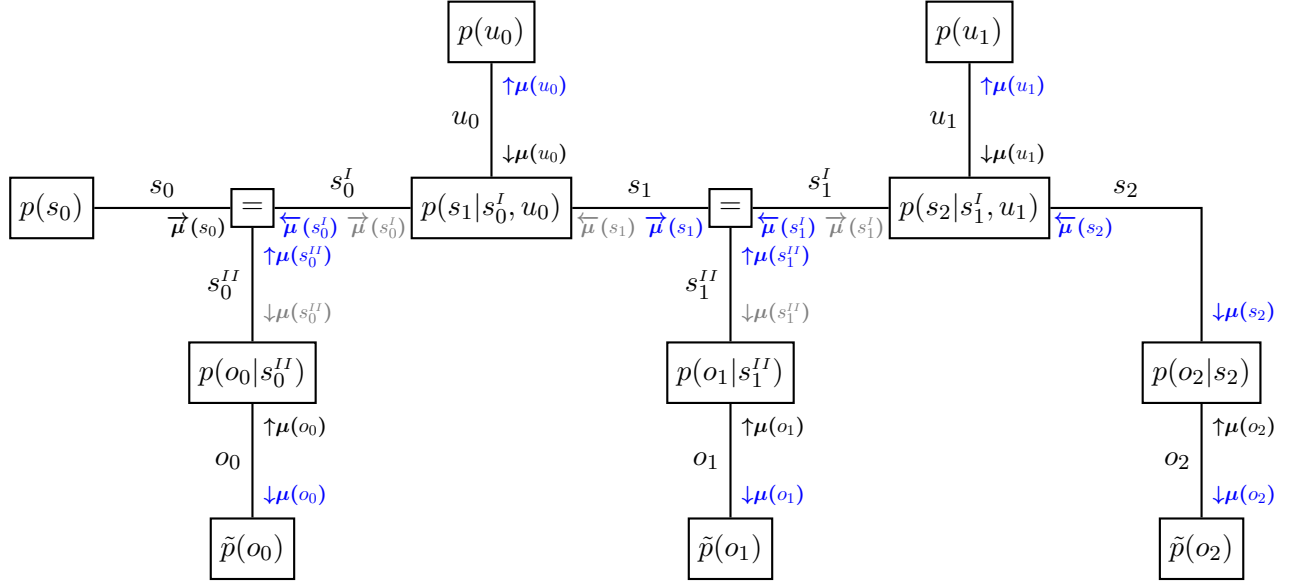


Figure 4.15: Factor graph of the temporal generative model with Bethe messages (blue).

of two. This is a technical problem because there are variables (edges) that occur in more than two factor function (nodes). As a remedy factor graphs support equality nodes that artificially duplicate the variables and enforce an equality relation between the duplicated variables. This relation is implemented in the factor function of these nodes as a product of Kronecker (or Dirac) delta functions. As an example consider the the factor graph in Fig. 4.16 where we are interested in $\overleftarrow{\mu}(x)$, $\overrightarrow{\mu}(x^I)$, and $\downarrow\mu(x^{II})$. The factor function of the equality node is given by

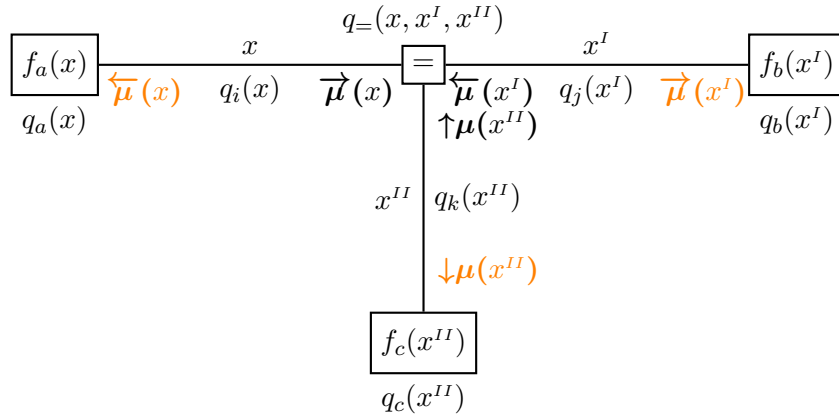


Figure 4.16: Example FFG with messages sent by the equality node (orange).

$$f = \delta_{x, x^I} \delta_{x, x^{II}}. \quad (4.101)$$

This function is a discrete probability distribution that has probability 1 only if the values of all variables are equal. In other cases it is 0. Building on this the node local free energy is

$$F_{=} = \sum_{x, x^I, x^{II}} q_{=}(x, x^I, x^{II}) \log \frac{q_{=}(x, x^I, x^{II})}{\delta_{x, x^I} \delta_{x, x^{II}}}. \quad (4.102)$$

To determine the messages next we need the free energy functional of the factor graph in Fig. 4.16 with

$$\begin{aligned} F = & \sum_{x, x^I, x^{II}} q_{=}(x, x^I, x^{II}) \log \frac{q_{=}(x, x^I, x^{II})}{\delta_{x, x^I} \delta_{x, x^{II}}} + \sum_x q_a(x) \log \frac{q_a(x)}{f_a(x)} + \sum_{x^I} q_b(x^I) \log \frac{q_b(x^I)}{f_b(x^I)} \\ & + \sum_z q_c(x^{II}) \log \frac{q_c(x^{II})}{f_c(x^{II})} + \sum_x q_i(x) \log \frac{1}{q_i(x)} + \sum_{x^I} q_j(x^I) \log \frac{1}{q_j(x^I)} \\ & + \sum_z q_k(x^{II}) \log \frac{1}{q_k(x^{II})}. \end{aligned} \quad (4.103)$$

Adding marginalization and normalization constraints to F leads to the Lagrangian

$$\begin{aligned} L = F + & \sum_x \lambda_{i=}(x) (q_i(x) - \sum_{x^I, x^{II}} q_{=}(x, x^I, x^{II})) + \sum_{x^I} \lambda_{j=}(x^I) (q_j(x^I) - \sum_{x, x^{II}} q_{=}(x, x^I, x^{II})) \\ & + \sum_{x^{II}} \lambda_{k=}(x^{II}) (q_k(x^{II}) - \sum_{x, x^I} q_{=}(x, x^I, x^{II})) + \sum_x \lambda_{ia}(x) (q_i(x) - q_a(x)) \\ & + \sum_{x^I} \lambda_{jb}(x^I) (q_j(x^I) - q_b(x^I)) + \sum_{x^{II}} \lambda_{kc}(x^{II}) (q_k(x^{II}) - q_c(x^{II})) \\ & + \psi_{=}(\sum_{x, x^I, x^{II}} q_{=}(x, x^I, x^{II}) - 1) + \psi_a(\sum_x q_a(x) - 1) + \psi_{x^I}(\sum_{x^I} q_b(x^I) - 1) \\ & + \psi_c(\sum_z q_c(x^{II}) - 1) + \psi_i(\sum_x q_i(x) - 1) + \psi_j(\sum_{x^I} q_j(x^I) - 1) + \psi_k(\sum_{x^{II}} q_k(x^{II}) - 1) \end{aligned} \quad (4.104)$$

Solving the belief distributions in L for the stationary points then yields

$$\frac{\delta L}{\delta q_{=}(x, x^I, x^{II})} = \log q_{=}(x, x^I, x^{II}) + 1 - \log(\delta_{x, x^I} \delta_{x, x^{II}}) - \lambda_{i=}(x) - \lambda_{j=}(x^I) - \lambda_{k=}(x^{II}) + \psi_{=} \stackrel{!}{=} 0 \quad (4.105)$$

$$\Rightarrow q_{=}^*(x, x^I, x^{II}) \propto \delta_{x, x^I} \delta_{x, x^{II}} \exp(\lambda_{i=}(x)) \exp(\lambda_{j=}(x^I)) \exp(\lambda_{k=}(x^{II})). \quad (4.106)$$

Similarly

$$\Rightarrow q_a^*(x) \propto f_a(x) \exp(\lambda_{ia}(x)) \quad (4.107)$$

$$\Rightarrow q_b^*(x^I) \propto f_b(x^I) \exp(\lambda_{jb}(x^I)) \quad (4.108)$$

$$\Rightarrow q_c^*(x^{II}) \propto f_c(x^{II}) \exp(\lambda_{kc}(x^{II})) \quad (4.109)$$

$$\Rightarrow q_i^*(x) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{i=}(x)) \quad (4.110)$$

$$\Rightarrow q_j^*(x^I) \propto \exp(\lambda_{j=}(x^I)) \exp(\lambda_{jb}(x^I)) \quad (4.111)$$

$$\Rightarrow q_k^*(x^{II}) \propto \exp(\lambda_{k=}(x^{II})) \exp(\lambda_{kc}(x^{II})). \quad (4.112)$$

Next, we use the marginalization constraint $\sum_{x^I, x^{II}} q_{=}(x, x^I, x^{II}) \stackrel{!}{=} q_i^*(x)$ to solve for the message sent by the equality node to node a (via edge i) with

$$\sum_{x^I, x^{II}} q_{=}(x, x^I, x^{II}) \stackrel{!}{=} q_i^*(x) \quad (4.113)$$

$$\sum_{x^I, x^{II}} \delta_{x, x^I} \delta_{x, x^{II}} \exp(\lambda_{i=}(x)) \exp(\lambda_{j=}(x^I)) \exp(\lambda_{k=}(x^{II})) \propto \exp(\lambda_{ia}(x)) \exp(\lambda_{i=}(x)) \quad (4.114)$$

$$\Rightarrow \exp(\lambda_{ia}(x)) \propto \sum_{x^I, x^{II}} \delta_{x, x^I} \delta_{x, x^{II}} \exp(\lambda_{j=}(x^I)) \exp(\lambda_{k=}(x^{II})) \quad (4.115)$$

$$\Rightarrow \overleftarrow{\mu}(x) \propto \sum_{x^I, x^{II}} \delta_{x, x^I} \delta_{x, x^{II}} \overleftarrow{\mu}(x^I) \uparrow \mu(x^{II}) \quad (4.116)$$

$$(\Rightarrow \overleftarrow{\mu}(x) \propto \overleftarrow{\mu}(x) \uparrow \mu(x)). \quad (4.117)$$

Analogously using the constraints $\sum_{x, x^{II}} q_{=}(x, x^I, x^{II}) \stackrel{!}{=} q_j^*(x^I)$ and $\sum_{x, x^I} q_{=}(x, x^I, x^{II}) \stackrel{!}{=} q_k^*(x^{II})$ we obtain

$$\Rightarrow \overrightarrow{\mu}(x^I) \propto \sum_{x, x^{II}} \delta_{x, x^I} \delta_{x, x^{II}} \overrightarrow{\mu}(x) \uparrow \mu(x^{II}) \quad (4.118)$$

$$\Rightarrow \downarrow \mu(x^{II}) \propto \sum_{x, x^I} \delta_{x, x^I} \delta_{x, x^{II}} \overrightarrow{\mu}(x) \overleftarrow{\mu}(x^I). \quad (4.119)$$

This reveals that outgoing messages from equality nodes (e.g., $\overleftarrow{\mu}(x)$ in Eq. (4.116)) are the product of incoming messages ($\overleftarrow{\mu}(x^I) \uparrow \mu(x^{II})$ in Eq. (4.116)) (except from the neighboring node that is the addressee of the message to be sent) where variables of incoming messages get substituted with the variable of the outgoing message as a result of the summation and the two Kronecker deltas. While in principle this substitution leads to Eq. (4.117) I refrain from that further step because the resulting messages ($\overleftarrow{\mu}(x) \uparrow \mu(x)$) might be misunderstood for the messages that are actually sent on the edge of variable x . Instead they really are the incoming messages from the edges of variables x^I and x^{II} respectively where the variables got substituted with x . Relating back to the previous section the messages sent by equality nodes are Bethe messages with the constraint of having a product of Kronecker deltas as the factor function.

Following this the messages sent by equality nodes in the factor graph of the temporal generative model (Fig. 4.17) can now be clearly stated with

$$\begin{aligned} \overleftarrow{\mu}(s_0) &= \sum_{s_0^I, s_0^{II}} \delta_{s_0, s_0^I} \delta_{s_0, s_0^{II}} \uparrow \mu(s_0^{II}) \overleftarrow{\mu}(s_0^I) & \overleftarrow{\mu}(s_1) &= \sum_{s_1^I, s_1^{II}} \delta_{s_1, s_1^I} \delta_{s_1, s_1^{II}} \uparrow \mu(s_1^{II}) \overleftarrow{\mu}(s_1^I) \\ (4.120) & & (4.123) \end{aligned}$$

$$\begin{aligned} \overrightarrow{\mu}(s_0^I) &= \sum_{s_0^I, s_0^{II}} \delta_{s_0, s_0^I} \delta_{s_0, s_0^{II}} \overrightarrow{\mu}(s_0) \uparrow \mu(s_0^{II}) & \overrightarrow{\mu}(s_1^I) &= \sum_{s_1^I, s_1^{II}} \delta_{s_1, s_1^I} \delta_{s_1, s_1^{II}} \overrightarrow{\mu}(s_1) \uparrow \mu(s_1^{II}) \\ (4.121) & & (4.124) \end{aligned}$$

$$\begin{aligned} \downarrow \mu(s_0^{II}) &= \sum_{s_0^I, s_0^{II}} \delta_{s_0, s_0^I} \delta_{s_0, s_0^{II}} \overrightarrow{\mu}(s_0) \overleftarrow{\mu}(s_0^I) & \downarrow \mu(s_1^{II}) &= \sum_{s_1^I, s_1^{II}} \delta_{s_1, s_1^I} \delta_{s_1, s_1^{II}} \overrightarrow{\mu}(s_1) \overleftarrow{\mu}(s_1^I) \\ (4.122) & & (4.125) \end{aligned}$$

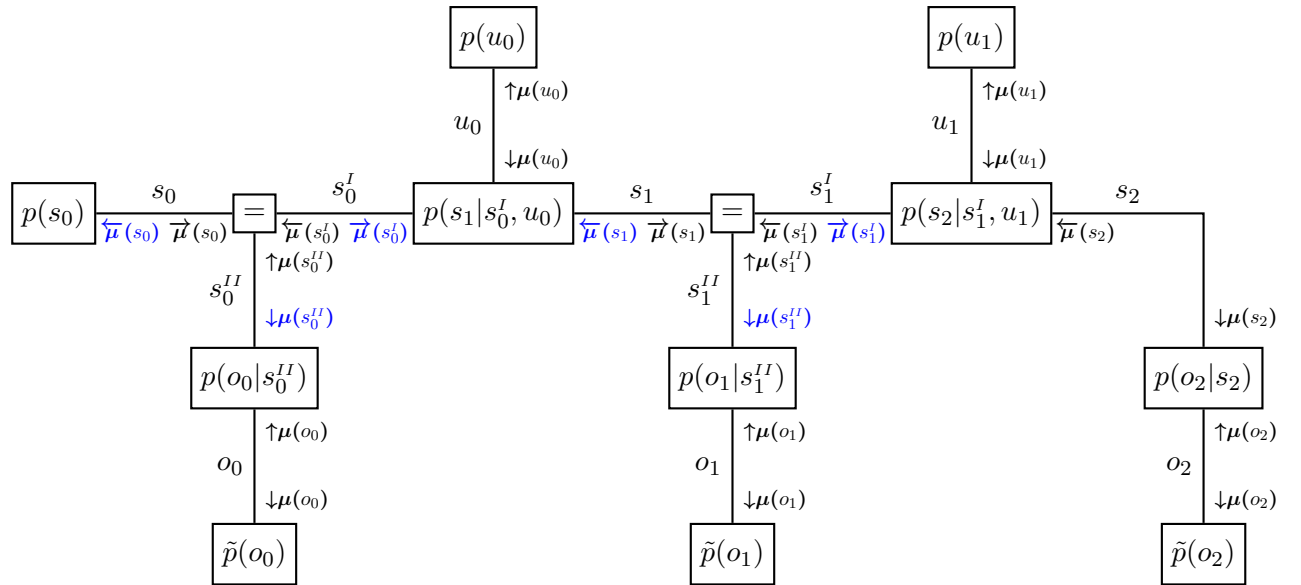


Figure 4.17: Factor graph of the temporal generative model with messages sent by equality nodes (blue).

Observation (Delta) Constraints

In this section I present how observations that are generated by the environment can be perceived by the agent by imposing constraints on the factor graph of its temporal generative model. Additionally this section also contains the derivation of messages resulting from these constraints in the graph. The agent explicitly models observations at different time points, e.g., with o_0 , o_1 , and o_2 as variables in the factor graph of its temporal generative model. In general, perception for the agent means that the

observation variable o_t of the current time step t gets fixed (clamped) to the actual observed value o'_t . The clamping is realized by constraining the corresponding belief distribution $q(o_t)$ of the current observation variable with a delta distribution δ_{o_t, o'_t} . This effectively puts all the probability mass at the event of the actual observed value o'_t . When doing inference, i.e., message passing in the factor graph these observed values persist and thus drive and influence the remaining "free" belief distributions (e.g., from control states) in the graph. I will elaborate on this mechanism later in this section. For now I will consider the example FFG in Fig. 4.18 that has a delta constraint and derive the messages ($\overleftarrow{\mu}(x)$, $\overrightarrow{\mu}(x)$) following the constraint.

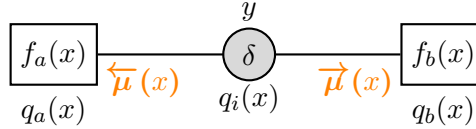


Figure 4.18: Example FFG with a delta constraint and the resulting messages (orange).

Having a delta constraint on the edge i is visualized with a gray bead in the middle of that edge. As mentioned before this implicates that the corresponding belief distribution q_i is a delta distribution that has its whole probability mass at a certain event x' , i.e., $q_i(x) = \delta_{x, x'}$. This constraint leads to specific messages $\overleftarrow{\mu}(x)$, $\overrightarrow{\mu}(x)$ as follows. For the derivation of these messages we follow the usual steps from previous sections and first define the free energy functional, then impose marginalization and normalization constraints to arrive at the Lagrangian and finally solve for the stationary solutions of the q distributions and rearrange for the messages using the marginalization constraints. The free energy functional of the factor graph in Fig. 4.18 is

$$F = \sum_x q_a(x) \log \frac{q_a(x)}{f_a(x)} + \sum_x q_b(x) \log \frac{q_b(x)}{f_b(x)} + \sum_x q_i(x) \log \frac{1}{q_i(x)}, \quad (4.126)$$

where the delta distribution is not yet substituted for $q_i(x)$. Following this, the Lagrangian then is

$$L = F + \psi_a \left(\sum_x q_a(x) - 1 \right) + \psi_b \left(\sum_x q_b(x) - 1 \right) + \psi_i \left(\sum_x q_i(x) - 1 \right) \\ + \sum_x \lambda_{ia}(x) (q_i(x) - q_a(x)) + \sum_x \lambda_{ib}(x) (q_i(x) - q_b(x)). \quad (4.127)$$

Solving the belief distributions in L for the stationary points then yields

$$\frac{\delta L}{\delta q_a(x)} = \log q_a(x) + 1 - \log f_a(x) - \lambda_{ia}(x) + \psi_b \stackrel{!}{=} 0 \quad (4.128)$$

$$\Rightarrow q_a^*(x) \propto f_a(x) \exp(\lambda_{ia}(x)). \quad (4.129)$$

Similarly

$$\Rightarrow q_b^*(x) \propto f_b(x) \exp(\lambda_{ib}(x)). \quad (4.130)$$

Solving for the stationary solution of $q_i(x)$ is not necessary because we already know that it is defined as a delta distribution, i.e.,

$$\Rightarrow q_i^*(x) \propto \delta_{x,x'}. \quad (4.131)$$

Next, we use the marginalization constraint $q_a^*(x) \stackrel{!}{=} q_i^*(x)$ to solve for the message $\overleftarrow{\mu}(x)$ with

$$q_a^*(x) \stackrel{!}{=} q_i^*(x) \quad (4.132)$$

$$f_a(x) \exp(\lambda_{ia}(x)) \propto \delta_{x,x'} \quad (4.133)$$

$$\Rightarrow \exp(\lambda_{ia}(x)) \propto \frac{\delta_{x,x'}}{f_a(x)} \quad (4.134)$$

$$\Rightarrow \exp(\lambda_{ia}(x)) \propto \delta_{x,x'} \quad (4.135)$$

$$\Rightarrow \overleftarrow{\mu}(x) \propto \delta_{x,x'}. \quad (4.136)$$

Note that $\frac{\delta_{x,x'}}{f_a(x)}$ in Eq. (4.134) evaluates to $\delta_{x,x'}$. To understand this, consider the following two cases. First, if $x \neq x'$ then both $\frac{\delta_{x,x'}}{f_a(x)}$ and $\delta_{x,x'}$ yield 0 as the result. Second, if $x = x'$ then $\delta_{x,x'}$ will result into 1 and $\frac{\delta_{x,x'}}{f_a(x)}$ will result into some positive number (not 0). For the latter normalization ensures that this arbitrary number will be 1 because all other outcomes already have probability 0 as outlined in the first case.

Similarly using the constraint $q_b^*(x) \stackrel{!}{=} q_i^*(y)$ we obtain

$$\Rightarrow \overrightarrow{\mu}(x) \propto \delta_{x,x'}. \quad (4.137)$$

The derivations hint that a delta constraint on an edge leads to messages that are equal to that delta distribution. In turn this means, that neither node a or b receive any information from the neighboring node because the delta constraint effectively blocks any information flow on that edge.

Now we can transfer the derived messages on edges with delta distributions to show how observations get integrated into the factor graph of the temporal generative model of the agent. For this we consider the scenario where the agent has already perceived an observation o'_0 in the past at timestep 0 and is currently at timestep 1 where it also got an observation o'_1 . The resulting factor graph with corresponding delta constraints is given in Fig. 4.19.

Following the blueprint from the example FFG in Fig. 4.18 the messages on the edges with delta constraints are

$$\uparrow\mu(o_0) \propto \delta_{o_0,o'_0} \quad (4.138) \quad \uparrow\mu(o_1) \propto \delta_{o_1,o'_1} \quad (4.140)$$

$$\downarrow\mu(o_0) \propto \delta_{o_0,o'_0} \quad (4.139) \quad \downarrow\mu(o_1) \propto \delta_{o_1,o'_1}. \quad (4.141)$$

In this situation the agent has effectively remembered what it has observed in the past at $t = 0$. This memory together with the observation at the current timestep $t = 1$

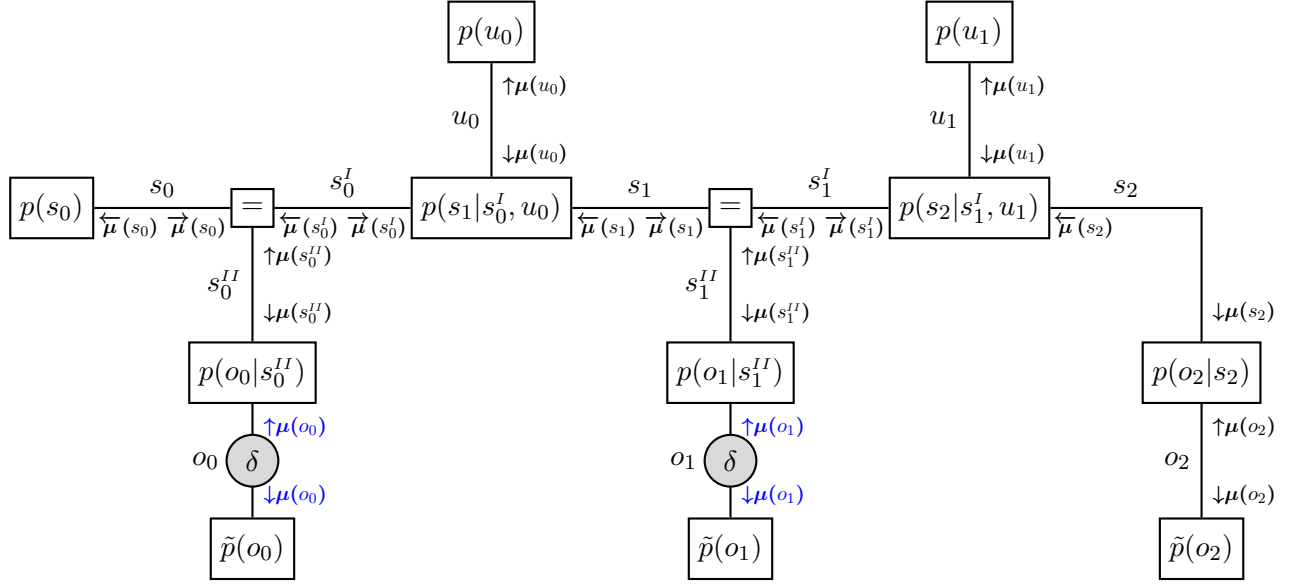


Figure 4.19: Factor graph of the temporal generative model with messages sent on edges with delta constraints (blue).

influences what the agent will infer for the belief distributions of the remaining variables in the temporal generative model. This is the central mechanism that enables the agent to form beliefs about future actions (e.g., u_1) that when executed lead to disered future observations (e.g., o_2) with respect to preferences encoded by $\tilde{p}(o_2)$. In other words information flows from past and current delta constraints (starting with messages $\uparrow\mu(o_0)$ and $\uparrow\mu(o_1)$) into parts of the temporal generative model that represents the future. This information flow in the factor graph realizes the planning process of the agent. Going further, when time proceeds to timestep $t = 2$ then another delta constraint gets applied to the edge of variable o_2 and both o_0 and o_1 are considered past observations, i.e., the memory of the agent. Also note that past preferences on observations (e.g., $\tilde{p}(o_0)$ and $\tilde{p}(o_1)$) get blocked by the delta constraints, i.e, they do not influence other belief distributions in the graph during the inference process anymore.

Action Selection

Belief Updating

Building on the results of the foundation section the edges' belief distributions are the product of the two messages of the respective edges. The edges' belief distributions in the factor graph of the temporal generative model (Fig. 4.20) are then given by

$$q(s_0) \propto \overleftarrow{\mu}(s_0) \overrightarrow{\mu}(s_0) \quad (4.142) \quad q(s_2) \propto \overleftarrow{\mu}(s_2) \downarrow \mu(s_2) \quad (4.148)$$

$$q(s_0^I) \propto \overleftarrow{\mu}(s_0^I) \overrightarrow{\mu}(s_0^I) \quad (4.143) \quad q(u_0) \propto \uparrow \mu(u_0) \downarrow \mu(u_0) \quad (4.149)$$

$$q(s_0^{II}) \propto \uparrow \mu(s_0^{II}) \downarrow \mu(s_0^{II}) \quad (4.144) \quad q(u_1) \propto \uparrow \mu(u_1) \downarrow \mu(u_1) \quad (4.150)$$

$$q(s_1) \propto \overleftarrow{\mu}(s_1) \overrightarrow{\mu}(s_1) \quad (4.145) \quad q(o_0) \propto \uparrow \mu(o_0) \downarrow \mu(o_0) \quad (4.151)$$

$$q(s_1^I) \propto \overleftarrow{\mu}(s_1^I) \overrightarrow{\mu}(s_1^I) \quad (4.146) \quad q(o_1) \propto \uparrow \mu(o_1) \downarrow \mu(o_1) \quad (4.152)$$

$$q(s_1^{II}) \propto \uparrow \mu(s_1^{II}) \downarrow \mu(s_1^{II}) \quad (4.147) \quad q(o_2) \propto \uparrow \mu(o_2) \downarrow \mu(o_2). \quad (4.153)$$

Note that we are referring here to the factor graph that already models $q(o_0)$ and $q(o_1)$ as delta distributions (fixed observations) which in principle makes the evaluation of Eq. (4.151) and Eq. (4.152) unnecessary.

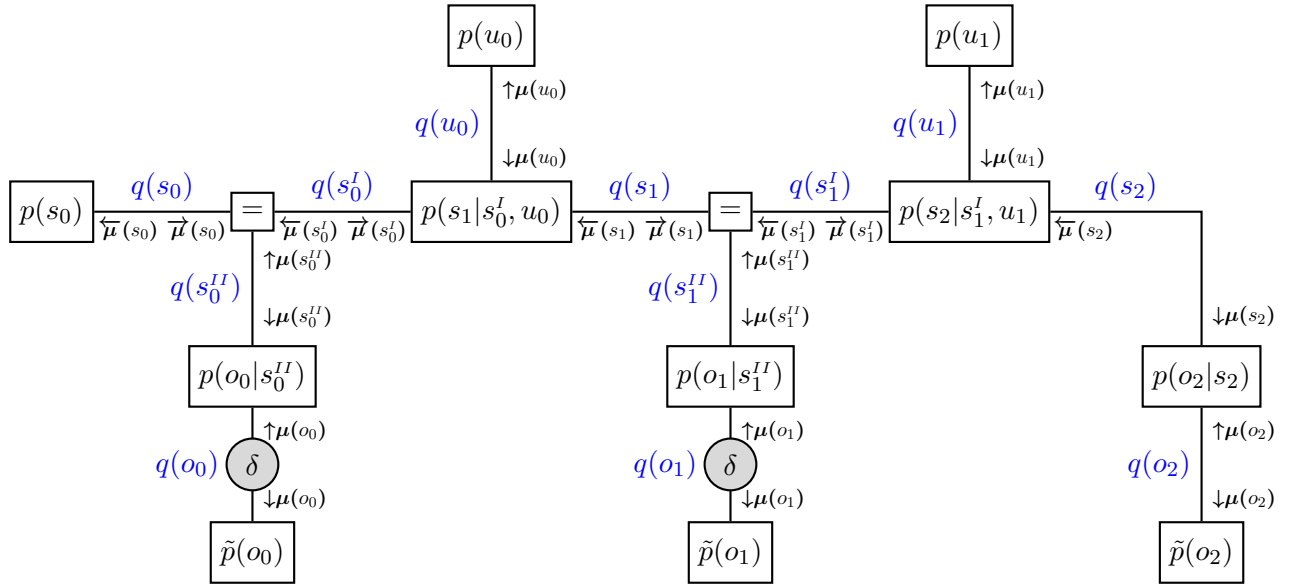


Figure 4.20: Factor graph of the temporal generative model with edges' belief distributions (blue).

Schedule

As mentioned in earlier sections the order of execution of messages plays an important role in the message passing scheme. In this section the agent largely relies on Bethe messages. In this case messages regularly depend on other messages which makes a correct schedule indispensable (later sections present other types of messages that allow for a lesser strict schedule). The schedule for the agent's temporal generative model in Fig. 4.20 is given in Algorithm 2.

Algorithm 2 Message Passing Algorithm of the BFE Agent

- 1: **evaluate** $\vec{\mu}(s_0), \uparrow\mu(o_0), \downarrow\mu(u_0), \uparrow\mu(o_1), \downarrow\mu(u_1), \uparrow\mu(o_2), \downarrow\mu(o_0), \downarrow\mu(o_1)$
 - 2: **evaluate** $\uparrow\mu(s_0^{II}), \uparrow\mu(s_1^{II}), \overleftarrow{\mu}(s_2)$
 - 3: **evaluate** $\vec{\mu}(s_0^I), \overleftarrow{\mu}(s_1^I)$
 - 4: **evaluate** $\vec{\mu}(s_1), \overleftarrow{\mu}(s_1)$
 - 5: **evaluate** $\vec{\mu}(s_1^I), \downarrow\mu(s_1^{II}), \uparrow\mu(u_0), \overleftarrow{\mu}(s_0^I)$
 - 6: **evaluate** $\uparrow\mu(u_1), \downarrow\mu(s_2), \overleftarrow{\mu}(s_0), \downarrow\mu(s_0^{II})$
 - 7: **evaluate** $\downarrow\mu(o_2)$
 - 8: **evaluate** $q(s_0), q(s_0^I), q(s_0^{II}), q(s_1), q(s_1^I), q(s_1^{II}), q(s_2), q(u_0), q(u_1), q(o_0), q(o_1), q(o_2)$
-

4.2.2 Generalized Free Energy (GFE) Agent

Generative Model

Variational Messages

GFE-Based Messages

Schedule

4.2.3 Context-BFE Agent

Generative Model

Structured Variational Messages

Schedule

4.2.4 Context-GFE Agent

Generative Model

GFE-Based Messages for the Context

Schedule

4.3 The T-Maze Environment

In this section I present the T-Maze environment, which all agents interact with during the simulations. Intuitively, the T-Maze is a maze in which an agent can move (through actions) and similarly make observations at the different locations in the maze. A schematic of the T-Maze is given in Fig. 4.21. The maze consists of four fields (Left Arm, Right Arm, Start, and Cue) and at every timestep the agent can be located at one of them (here the agent is at the start field). The agent has four different actions that can move it to the four possible fields in the maze. The usual interaction between the agent and the T-Maze during a simulation is as follows.

The agent always begins at the start field. Moreover, the agent has a (strong) preference for observing the reward, which is located either at the left arm or the right arm. Unfortunately, the agent does initially not know at which arm the reward is. Following this, it has two options to find the reward. First, it could randomly move to one of the two arms and receive the reward with 50 % chance. Second, it could first move to the cue, which gives away information about the reward location (including uncertainty). Using this knowledge the agent can then make a better decision when moving to one of the two arms. In summary, the simulation lasts at maximum two timesteps during which the agent can either directly move to one of the arms (which ends the simulation after the first timestep) or it can first move to the cue and then move to one of the arms.

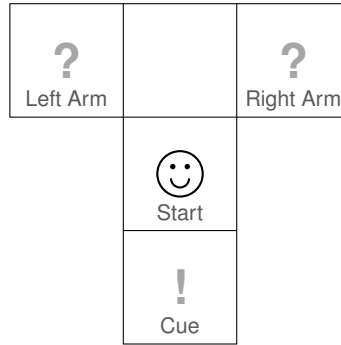


Figure 4.21: The figure shows the T-Maze environment with its four fields (Left Arm, Right Arm, Start, and Cue) and the agent (Smiley) that is currently located at the start field. The empty space above the start field is not considered a field because the agent cannot move there.

The T-Maze environment is precisely described by a corresponding generative process that relates actions from the agent, hidden environmental states and observations generated by the environment probabilistically. Specifically, observations o are generated by (depend on) hidden environmental states s via the likelihood distribution $p(o|s)$. Hidden environmental states evolve over time (from t to $t + 1$) based on actions a via $p(s_{t+1}|s_t, a_t)$. At timestep $t = 0$ hidden environmental states are given by the prior $p(s_0)$. Before giving the probabilities of these relationships, I first define the observations, hidden states, and actions. Specifically, observations and hidden states can be defined as a cartesian product of three other sets. These sets are

$$\text{Field} = \{L, R, S, C\}, \quad (4.154)$$

that defines the four fields (L (Left Arm), R (Right Arm), S (Start), C (Cue)) in the T-Maze,

$$\text{MazeType} = \{A, B\}, \quad (4.155)$$

that defines the two possible types of the T-Maze, and

$$\text{Outcome} = \{RW, NR, CL, CR\}, \quad (4.156)$$

that defines the four possible outcomes (not to be confused with observations) at a field (RW (Reward), NR (No Reward), CL (Cue hints at Left arm), CR (Cue hints at Right arm)). Observations $o \in O$ are defined as

$$\begin{aligned} O &= \text{Field} \times \text{Outcome} \\ &= \{(L, RW), (L, NR), (L, CL), (L, CR), (R, RW), (R, NR), (R, CL), (R, CR), \\ &\quad (S, RW), (S, NR), (S, CL), (S, CR), (C, RW), (C, NR), (C, CL), (C, CR)\}, \end{aligned}$$

hidden states $s \in S$ are given by

$$\begin{aligned} S &= \text{Field} \times \text{MazeType} \\ &= \{(L, A), (L, B), (R, A), (R, B), (S, A), (S, B), (C, A), (C, B)\}, \end{aligned}$$

and actions $a \in A$ are

$$A = \{\text{GtL}, \text{GtR}, \text{GtS}, \text{GtC}\},$$

meaning GtL (Go to Left arm), GtR (Go to Right arm), GtS (Go to Start), and GtC (Go to Cue). Next, I detail the probabilistic relationships between the variables. Observations are generated depending on hidden states via $p(o|s)$ as

$$\begin{aligned} p(o = (S, CL) \mid s = (S, A)) &= p(o = (S, CL) \mid s = (S, B)) = 0.5 \\ p(o = (S, CR) \mid s = (S, A)) &= p(o = (S, CR) \mid s = (S, B)) = 0.5 \\ p(o = (C, CL) \mid s = (C, A)) &= p(o = (C, CR) \mid s = (C, B)) = 1 \\ p(o = (L, RW) \mid s = (L, A)) &= p(o = (L, NR) \mid s = (L, B)) = \alpha \\ p(o = (R, RW) \mid s = (R, B)) &= p(o = (R, NR) \mid s = (R, A)) = \alpha \\ p(o = (L, NR) \mid s = (L, B)) &= p(o = (L, RW) \mid s = (L, A)) = 1 - \alpha \\ p(o = (R, NR) \mid s = (R, B)) &= p(o = (R, RW) \mid s = (R, A)) = 1 - \alpha \end{aligned}$$

where α is a parameter that gets modulated in the simulations and other probabilities $p(o|s)$ are zero. Hidden states $s_{k+1} \in S$ at time $k+1$ are generated depending on $s_k \in S$

and actions $a_k \in A$ at timestep k with probabilities

$$\begin{aligned} p(s_{k+1} = (S, A) \mid s_k = (S, A), a_k = \text{GtS}) &= p(s_{k+1} = (S, B) \mid s_k = (S, B), a_k = \text{GtS}) = 1 \\ p(s_{k+1} = (C, A) \mid s_k = (S, A), a_k = \text{GtC}) &= p(s_{k+1} = (C, B) \mid s_k = (S, B), a_k = \text{GtC}) = 1 \\ p(s_{k+1} = (L, A) \mid s_k = (S, A), a_k = \text{GtL}) &= p(s_{k+1} = (L, B) \mid s_k = (S, B), a_k = \text{GtL}) = 1 \\ p(s_{k+1} = (R, A) \mid s_k = (S, A), a_k = \text{GtR}) &= p(s_{k+1} = (R, B) \mid s_k = (S, B), a_k = \text{GtR}) = 1 \end{aligned}$$

$$\begin{aligned} p(s_{k+1} = (S, A) \mid s_k = (C, A), a_k = \text{GtS}) &= p(s_{k+1} = (S, B) \mid s_k = (C, B), a_k = \text{GtS}) = 1 \\ p(s_{k+1} = (C, A) \mid s_k = (C, A), a_k = \text{GtC}) &= p(s_{k+1} = (C, B) \mid s_k = (C, B), a_k = \text{GtC}) = 1 \\ p(s_{k+1} = (L, A) \mid s_k = (C, A), a_k = \text{GtL}) &= p(s_{k+1} = (L, B) \mid s_k = (C, B), a_k = \text{GtL}) = 1 \\ p(s_{k+1} = (R, A) \mid s_k = (C, A), a_k = \text{GtR}) &= p(s_{k+1} = (R, B) \mid s_k = (C, B), a_k = \text{GtR}) = 1 \end{aligned}$$

$$\begin{aligned} p(s_{k+1} = (S, A) \mid s_k = (L, A), a_k = \text{GtS}) &= p(s_{k+1} = (S, B) \mid s_k = (L, B), a_k = \text{GtS}) = 1 \\ p(s_{k+1} = (S, A) \mid s_k = (L, A), a_k = \text{GtC}) &= p(s_{k+1} = (S, B) \mid s_k = (L, B), a_k = \text{GtC}) = 1 \\ p(s_{k+1} = (S, A) \mid s_k = (L, A), a_k = \text{GtL}) &= p(s_{k+1} = (S, B) \mid s_k = (L, B), a_k = \text{GtL}) = 1 \\ p(s_{k+1} = (S, A) \mid s_k = (L, A), a_k = \text{GtR}) &= p(s_{k+1} = (S, B) \mid s_k = (L, B), a_k = \text{GtR}) = 1 \end{aligned}$$

$$\begin{aligned} p(s_{k+1} = (S, A) \mid s_k = (R, A), a_k = \text{GtS}) &= p(s_{k+1} = (S, B) \mid s_k = (R, B), a_k = \text{GtS}) = 1 \\ p(s_{k+1} = (S, A) \mid s_k = (R, A), a_k = \text{GtC}) &= p(s_{k+1} = (S, B) \mid s_k = (R, B), a_k = \text{GtC}) = 1 \\ p(s_{k+1} = (S, A) \mid s_k = (R, A), a_k = \text{GtL}) &= p(s_{k+1} = (S, B) \mid s_k = (R, B), a_k = \text{GtL}) = 1 \\ p(s_{k+1} = (S, A) \mid s_k = (R, A), a_k = \text{GtR}) &= p(s_{k+1} = (S, B) \mid s_k = (R, B), a_k = \text{GtR}) = 1 \end{aligned}$$

and other probabilities $p(s_{k+1} \mid s_k, a_k)$ are zero and initial hidden states at timestep $k = 0$ are defined via $p(s_0)$ as

$$p(s_0 = (S, A)) = p(s_0 = (S, B)) = 0.5$$

and other probabilities $p(o_0)$ are zero.

4.4 Simulations

This section presents the schedule of the three simulations ("Initial Curiosity", "Exploiting Information", and "Long-Term Success"). Each simulation is run for each agent type ("BFE", "Context-BFE", "GFE", and "Context-GFE") and parameters $\alpha \in [0.0, 0.1, \dots, 1.0]$ and $c \in [0, 1, \dots, 10]$ of their generative models. The simulations reveal different aspects of the agents' behavior. All simulations have in common that the agent begins in the start field of the T-Maze where it also initially receives an observation. This circumstance requires further explanation. First, the agent has a prior over its initial position in the T-Maze. This prior already entails the information that the agent begins at the

start field, i.e., the agent already knows that it is located at this start field. Following this, the observation at this field does not yield any new information for the agent—the observation reveals to the agents its current location which the agent already "knows". Furthermore, this initial observation then has no effect on the agent, i.e., inferred actions are not dependent on this observation.

4.4.1 Simulation 1: Initial Curiosity

This simulation is designed to show the agents' initial drive to visit the cue location in the T-Maze. This is not to be understood as an empirical test, i.e., we do not have a cohort of the same agent type and average their behaviors. Instead, for each agent type the simulation is run once and their inclination of initially visiting the cue can directly be read of from their action specific belief distribution in the current timestep. Concretely, the respective agent first gets exposed to the environment (T-Maze) where it is initially located at the start position and the environment generates an observation that the agent perceives. Next, the agent performs inference, i.e., its message passing algorithm is executed until convergence. As a result, the agent has inferred all of its belief distributions including the one that prescribes action in the current timestep. This action specific belief distribution contains probabilities over the four possible actions (go to (stay at) start, go to the left arm, go to the right arm, go to the cue). The agent's inclination of going to the cue is then exactly the respective probability of performing this action. This probability is the central result of this simulation. The procedere is exemplary detailed using the BFE agent in Fig. 4.22.

4.4.2 Simulation 2: Exploiting Information

In this simulation, the agent receives the cue which reveals the location of the reward (left arm or right arm) in the T-Maze. The result of this experiment is then the agent's inclination of going to the respective arm in the T-Maze that the cue indicates as given by the probabilities in its action-specific belief distribution. The simulation spans over timesteps $t = 0$ and $t = 1$. First, the agent receives an observation o_0 generated by the environment (at $t = 0$). Second, it executes the "go to cue" action and thus moves to the cue field in the environment, i.e., the action is not sampled from the action belief distribution $q(u_0)$ but is externally triggered (at $t = 0$). Third, it receives the second observation o_1 while being at the cue field (at $t = 1$). Fourth, the agent runs its messages passing algorithm to infer its belief distributions. After inference, the result of the experiment is given as the agent's belief (a probability) of executing the specific action that moves it to the arm that is indicated by the cue. A detailed explanation of the procedere is given in Fig. 4.23.

4.4.3 Simulation 3: Long-Term Success

This simulation is about empirically measuring the agents' succes in receiving the reward during a trial. Specifically, there are two ways how an agent can achieve this: First,

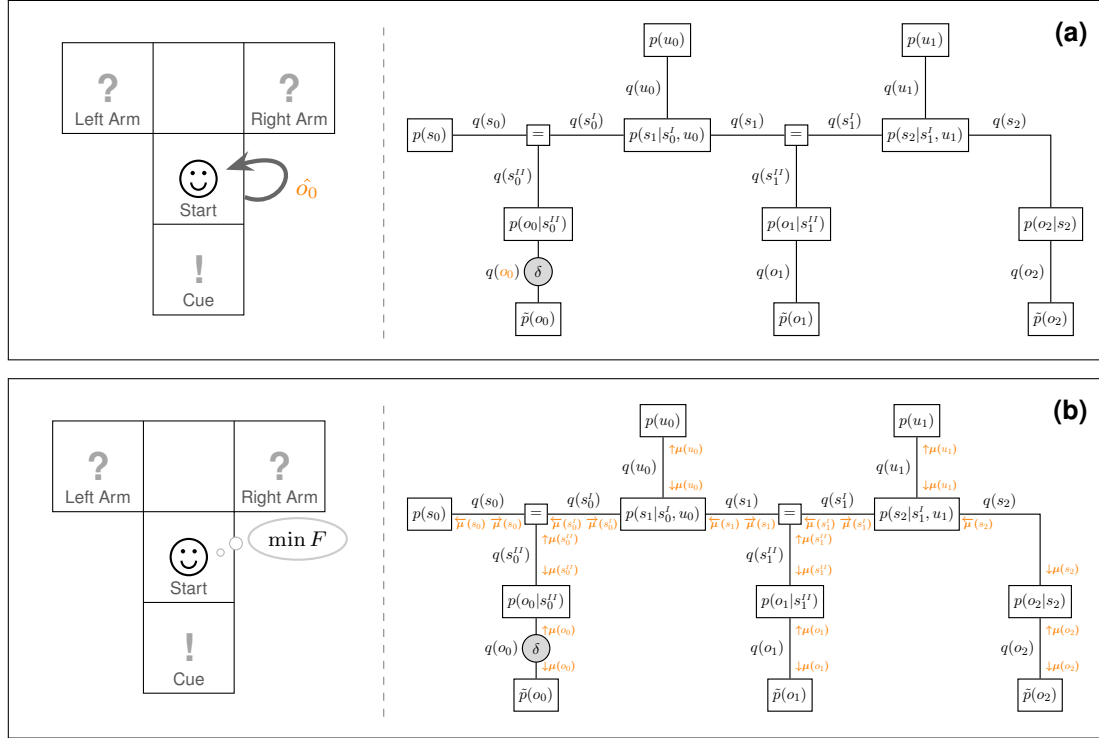
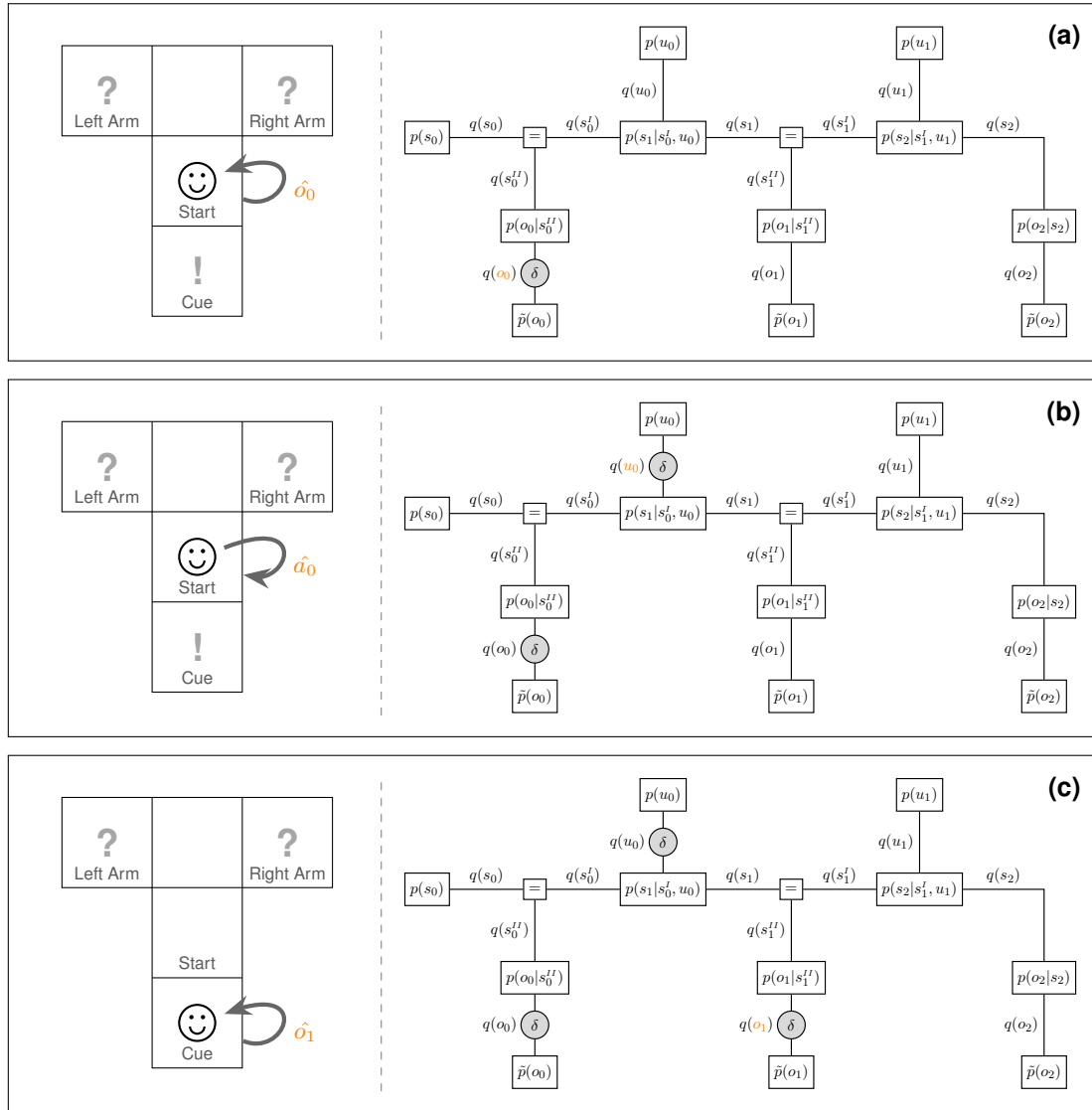


Figure 4.22: The figure presents the two crucial steps of the "Initial Curiosity" simulation using the example of the BFE agent: First, the agent receives an observation $\hat{o}_0$ from the environment (Fig. 4.22a). The observation has an effect on the agent's temporal generative model, i.e., in its factor graph representation, the belief distribution $q(o_0)$ that describes its belief about observations of timestep $t = 0$, becomes a delta distribution with all probability mass at $\hat{o}_0$. Second, the agent performs inference, i.e., it runs its message passing algorithm (messages sent along the edges in Fig. 4.22b). In the case of the BFE agent this refers to Algorithm 2. After message passing has finished, the beliefs of executing a certain action in the current timestep are then directly given by the belief $q(u_0)$.

it initially "decides" against going to the cue and instead moves to one of the two arms. It is then either lucky and observes the reward or not. Second, the agent initially moves to the cue and only then moves to one of the two arms. Again, it then either observes the reward or not. Generally, a trial consists of three timesteps $t \in 0, 1, 2$ and the agent can either observe the reward in timestep $t = 1$ (when directly moving to an arm) or in timestep $t = 2$ (when first moving to the cue in $t = 1$). A trial is succesfull when the agent observes the reward in timesteps $t = 1$ or $t = 2$ (a trial is prematurely terminated when it receives in reward in timestep $t = 1$). There are 1000 trials for each agent (and α - and c value). The result is the percentage of succesfull trials. The procedere is detailed in Fig. 4.24.



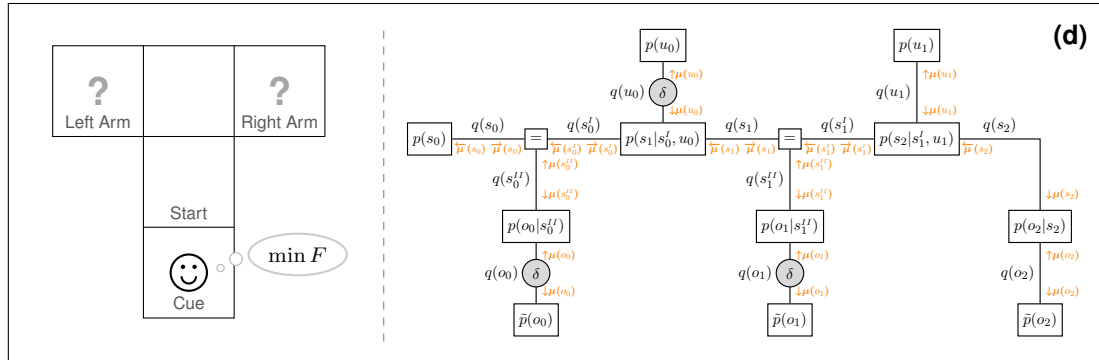
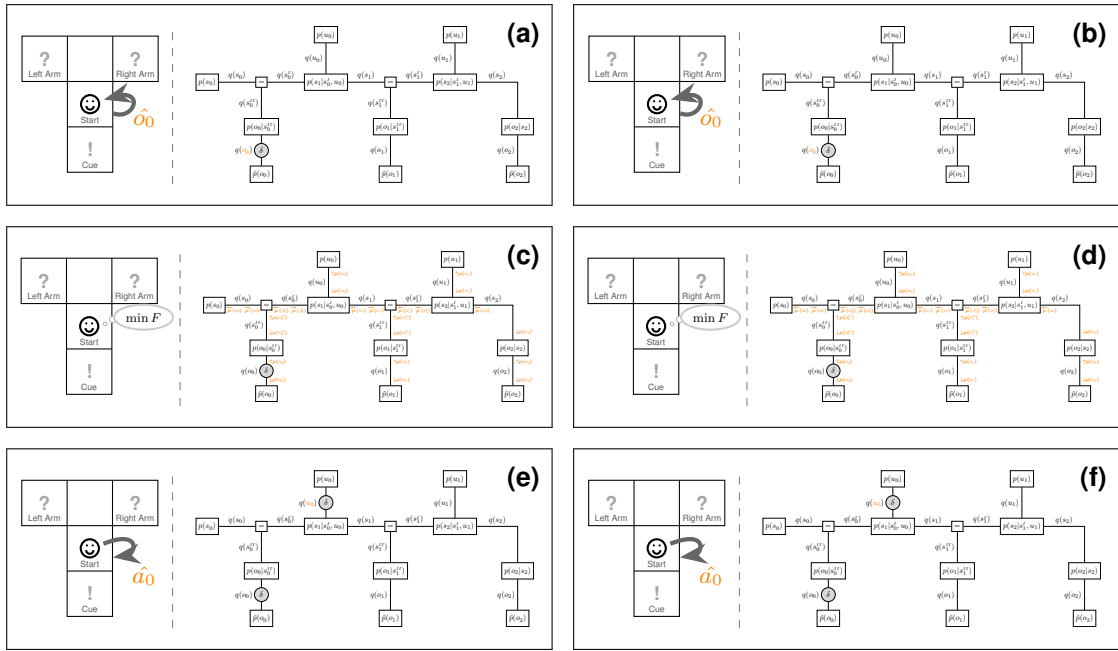


Figure 4.23: The figure presents the four steps of the simulation "Exploiting Information" via the example of the BFE agent. First, the agents receives an observation $\hat{o}_0$ which makes $q(o_0)$ a delta distribution with all its probability mass at $\hat{o}_0$ (Fig. 4.23a). Second, the action "go to cue" is externally triggered and the respective belief distribution $q(u_0)$ also becomes a delta distribution with its mass at this action (Fig. 4.23b). Third, being at the cue triggers an observation from the environment that is informative regarding the reward location. Similar to the first observation, it leads to a delta distribution at $q(o_1)$ (Fig. 4.23c). Lastly, the agent executes its message passing algorithm to infer all its belief distributions (Fig. 4.23d). The inclination of moving to the cued arm that promises the reward is then given as a probability in the action-specific belief distribution $q(u_1)$.



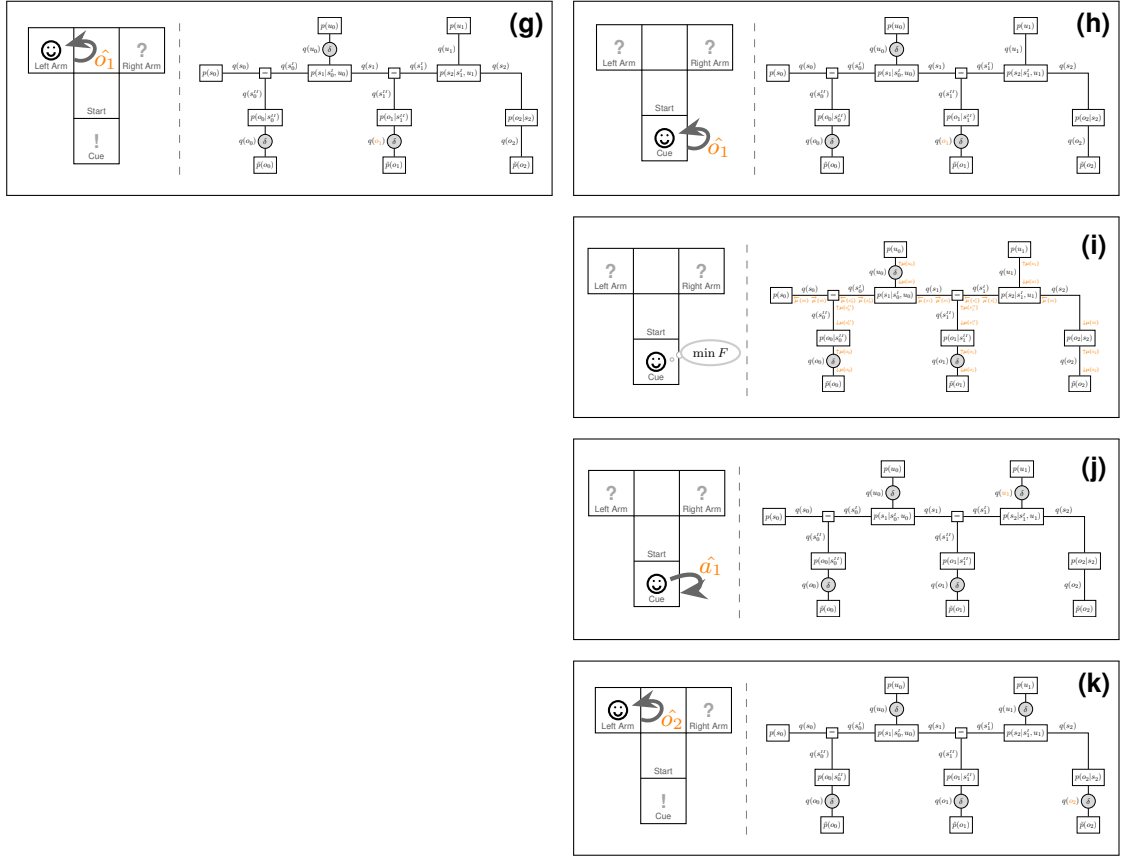


Figure 4.24: The figure shows the two behavioral paths the agent can take during a trial (left column and right column). In the left column the agent decides against moving to the cue and instead directly moves to one of the two arms. In the right column the agent first moves to the cue and then decides for one of the two arms. Both paths start similarly. Specifically, the agent always begins at the start field in the T-Maze and receives an observation (Figs. 4.24a and 4.24b). Next, the agent infers its belief distributions (Figs. 4.24c and 4.24d). Based on the inferred action-specific belief distribution the agent executes an action (Figs. 4.24e and 4.24f) where this action either moves it to the cue (Fig. 4.24h) or directly to one of the arms (Fig. 4.24g). Directly moving to one of the arms ends the trial (left column). Next, the agent in the right column again infers its belief distributions also taking into account the information received by the cue (Fig. 4.24i). Following this second inference, the agent can again act again ((Fig. 4.24j)) to finally move to one of the two arm ((Fig. 4.24k)). Note that in principle there are more behavioral paths than the two shown here but these other paths do never lead to the reward, e.g., always taking actions that keeps the agent at the start field. The two paths shown here are the only paths that have non-zero probabilities of leading to the reward, i.e., also with the two paths the agent might not receive the reward when moving to the wrong arm.

5 Results

In this section I present the results of the three experiments (Initial Curiosity, Exploiting Information, and Long-Term Success) detailed in Section 4.4. For each experiment the analyses are available for all four agents (BFE, GFE, Context-BFE, and Context-GFE) with two parameter sweeps ($\alpha \in [0.0, 0.1, \dots, 1.0]$, and $c \in [0, 1, \dots, 10]$) each. This leads to visualizations that contain four heatmaps (for the four agents) where the two axes represent the two parameter sweeps (over α and c). The result for a specific constellation of agent type, α value, and c value is then given in the specific cell of the heatmap. The heatmaps assign brighter and darker colors to higher and lower values respectively. For the following it is useful to recall what the different agents types, α values, and c values refer to.

Both BFE Agent and Context-BFE Agent are not curious and do not seek out information in the environment to reduce uncertainties in their generative models. Conversely, the GFE Agent and Context-GFE Agent do exactly that, i.e., they are indeed curious and are driven to reduce uncertainties. Orthogonally to the two properties "curious" and "non-curious", the agents can also be categorized as having a context state or not having a context state. Following this, both Context-BFE Agent and Context-GFE Agent possess a context state, which in the case of the Context-GFE Agent is subject to reducing its uncertainty as a result of the curiosity. Conversely, the BFE Agent and GFE Agent do not possess a context state and the curiosity of the GFE Agent then attempts to reduce uncertainty of the basic hidden state (all four agents possess hidden states as part of their generative models but two of them additionally have context states).

The α values are parameters of the agents' generative models. Using this model the agent can form a belief about the kind of T-Maze it is interacting with: Is it the T-Maze that has the reward in the left arm or is it the other T-Maze that has the reward in the right arm? That means the agent knows there are two types of T-Mazes with differing reward locations. Counterintuitively, labeling the two possible T-Mazes in this way (reward in left arm or reward in right arm) does not necessarily imply that the reward truly is at the referred location. This circumstance is depicted by the generative model by the parameter α . Values close to 1.0 mean that the agent can trust its belief about the reward location, e.g., when it assumes it is interacting with the T-Maze that has the reward in the left arm then it can be sure it will actually observe the reward in the left arm. Values close to 0.0 also mean that the agent can trust its belief about the reward location but in an inverted way, i.e., when it assumes it is interacting with the T-Maze that has the reward in the left arm then it can be sure it will actually observe the reward in the opposite (right) arm. Values near 0.5 implicate that knowing about the type of T-Maze (reward in left arm or reward in right arm) does (almost) not yield any information about the true reward location.

The c values are also parameters of the agents' generative models. They define how much the agent is actually interested in receiving (observing) the reward. To give an intuition for the values, $c = 0$ means it wants to receive the reward in 25% of the cases, $c = 1$ means it wants it in $\sim 53.4\%$ of the cases, $c = 2$ means it wants it in $\sim 77.6\%$ of the cases, $c = 5$ means it wants it in $\sim 98.7\%$ of the cases, and $c = 10$ means it wants it in $\sim 100\%$ of the cases.

5.1 Simulation 1: Initial Curiosity

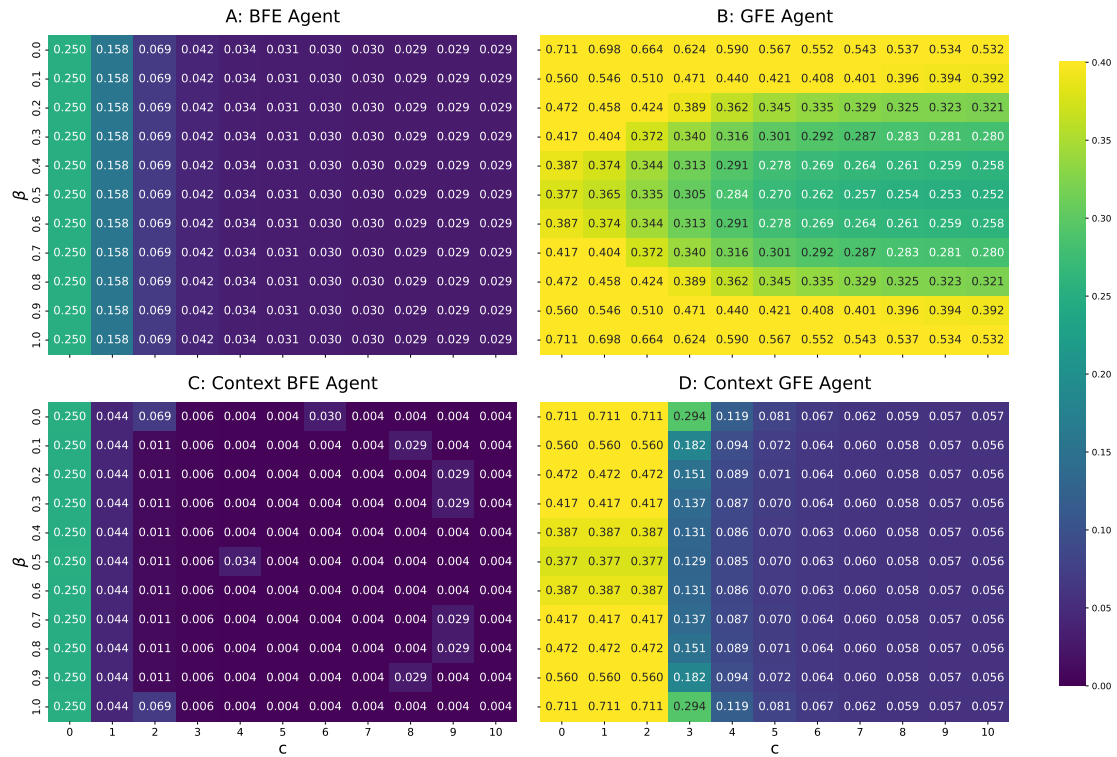


Figure 5.1: Inferred probabilities of executing an action in timestep $t = 0$ that moves the agent to the cue for the four agents (BFE, GFE, Context-BFE, and Context-GFE) with differing α - and c values.

The values in the heatmaps' cells (Fig. 5.1) give the inferred probabilities of executing the action in timestep $t = 0$ that moves the agent to the cue location in the T-Maze. The values are not based on empirical measurements but directly represent the agents' beliefs about their actions.

Fundamentally, the four plots reveal an axial symmetry around $\alpha = 0.5$, i.e., all four agents do not differentiate between a value α and its inverse $1 - \alpha$. That means both alpha values lead to the same inclination to visit the cue location in the T-Maze.

The results indicate a behavioral divide between the agents that are not curious (BFE and Context-BFE) and the ones that are curious (GFE and Context-GFE). In the former case (Fig. 5.1A and Fig. 5.1C) both results yield equal probabilities for all α - and c values, which means that both BFE- and Context-BFE agents show exactly the same behavior of going to the cue in the first timestep. For the latter (Fig. 5.1B and Fig. 5.1D) the results are not equal but the visual pattern regarding α - and c values is similar (tapering funnel of values marked with yellow colors with increasing c values), which generally means that also both GFE- and Context-GFE agents have similar tendencies for visiting the cue for differing α and c values.

Both non-curious agents (Fig. 5.1A and Fig. 5.1C) have invariant probabilities for different α values, which means that α has no effect on their inclination of visiting the cue. In contrast, increasing c values also lead to smaller probabilities starting with $p = 0.25$ for $c = 0$ and reaching $p = 0.167$ for $c = 10$, i.e., when the agent has stronger preferences for observing the reward it becomes less likely to initially visit the cue.

In the case of the curious agents (Fig. 5.1B and Fig. 5.1D) the results are more nuanced. Generally, extreme α values (close to 0.0 or 1.0) yield smaller probabilities compared to α values near 0.5 (e.g., $p = 0.362$ at $(\alpha = 0.0, c = 0)$ vs. $p = 0.380$ at $(\alpha = 0.5, c = 0)$ in Fig. 5.1B), i.e., being more certain about the true reward location upon knowing the type of T-Maze (reward in left arm or reward in right arm) makes the cue less interesting to visit. Furthermore, increasing c values monotonically decrease the probabilities (e.g., $p = 0.362$ at $(\alpha = 0.0, c = 0)$ to $p = 0.312$ at $(\alpha = 0.0, c = 10)$ in Fig. 5.1B) except for the cases when α is also close (or equal) to 0.5 (e.g., $p = 0.380$ at $\alpha = 0.5$ in Fig. 5.1B), i.e., similarly to the non-curious agents stronger preferences for observing the reward makes the agent less likely to visit the cue, except for the cases when knowing the type of environment does not yield a lot of information about the true reward location. In these cases the probabilities are invariant with respect to c . While the general behavioral pattern of both the GFE- and Context-GFE agent is similar there are differences, which are detailed in the following.

The Context-GFE agent yields smaller probabilities for increasing c values compared to the GFE agent (e.g., $p = 0.251$ at $(\alpha = 0.0, c = 10)$ in Fig. 5.1B vs. $p = 0.312$ at $(\alpha = 0.0, c = 10)$ in Fig. 5.1D)), except for α values close (or equal) to 0.5. Moreover, the two figures Fig. 5.1B and D visually reveal a smoother shift of probabilities from the regime of small c values and arbitrary α values to the regime of large c values and extreme α values (close to 0.0 or 1.0) for the GFE agent compared to the Context-GFE agent (e.g., $p = 0.348$ at $(\alpha = 0.3, c = 5)$ to $p = 0.328$ at $(\alpha = 0.2, c = 6)$ in Fig. 5.1B vs. $p = 0.378$ at $(\alpha = 0.3, c = 5)$ to $p = 0.274$ at $(\alpha = 0.2, c = 6)$ in Fig. 5.1D), i.e., the Context-GFE agent changes its behavior (inclination of going to the cue) more abruptly than the GFE agent.

5.2 Simulation 2: Exploiting Information

The values in the heatmaps' cells (Fig. 5.2) give the inferred probabilities of moving to the arm in the T-Maze that is hinted by the cue to contain the reward. Again, the

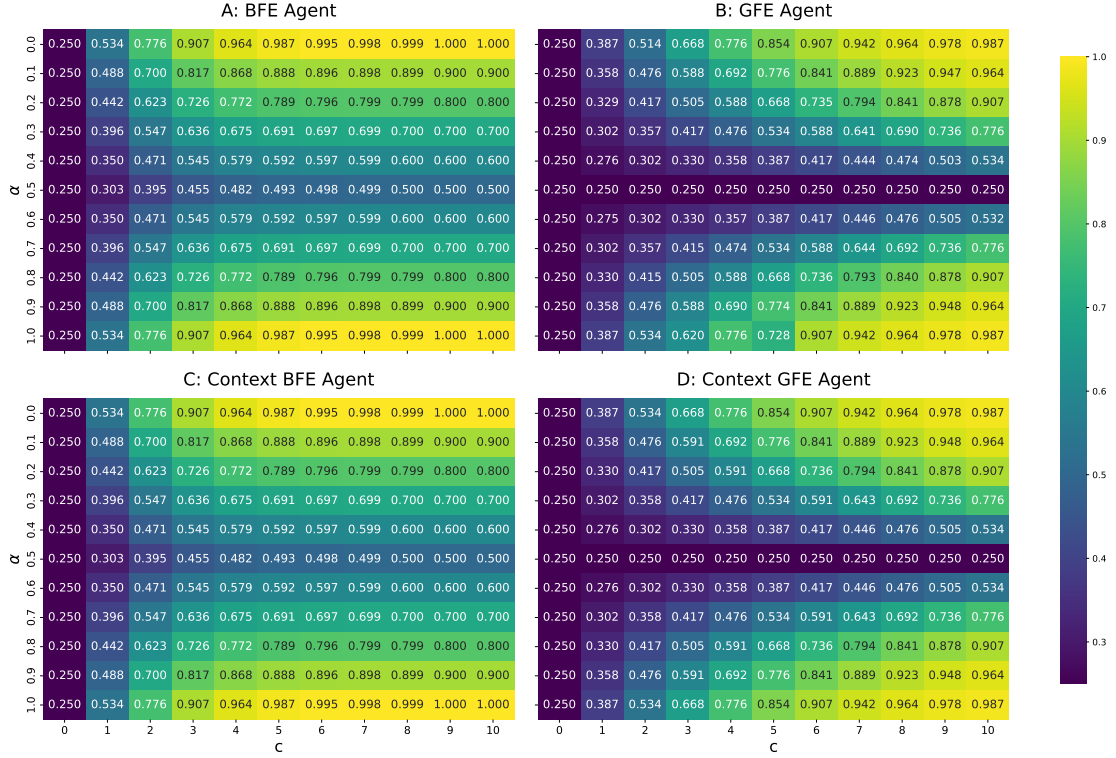


Figure 5.2: Inferred probabilities of moving to the arm in the T-Maze that is hinted by the cue to contain the reward for the four agents (BFE, GFE, Context-BFE, and Context-GFE) with differing α - and c values.

probabilities do not represent empirical measurements but beliefs of the agent that it will execute the action that moves it to the arm that should contain the reward according to the cue.

The four plots also reveal an axial symmetry around $\alpha = 0.5$. That means both alpha values (α and $1 - \alpha$) lead to the same inclination to visit the specific arm in the T-Maze that is hinted to contain the reward by the cue.

Similarly to the experiment "Initial Curiosity" we can observe a behavioral divide between the two non-curious agents (Fig. 5.2A and Fig. 5.2C) and the two curious agents (Fig. 5.2B and Fig. 5.2D), albeit not as drastically. Again, the BFE- and Context-BFE agents yield exactly the same probabilities. This behavioral equality is now also evident for the GFE- and Context-GFE agents. The subtle differences between non-curious and curious agents are given in the following.

Compared to the non-curious agents, the curious agents yield invariant probabilities for varying c values when α is fixed to 0.5 (i.e., $p = 0.250$ at $\alpha = 0.5$ in Fig. 5.2B vs. $p = 0.250$ to $p = 0.500$ at $\alpha = 0.5$ in Fig. 5.2A), i.e., compared to the non-curious agents the curious agent are not stronger inclined to visit the cued reward location with increasing preference for the reward when they "know" there is no correlation between

assumed T-Maze type (that is hinted by the cue) and the real reward location as given by $\alpha = 0.5$. Moreover, the probabilities of the curious agents ascend slower with increasing c values compared to the non-curious agents (e.g., $p = 0.9$ is surpassed at $(\alpha = 0.0, c = 6)$ in Fig. 5.2B vs. $p = 0.9$ is surpassed at $(\alpha = 0.0, c = 3)$ in Fig. 5.2A), i.e., the curious agents do not visit the cued reward location as frequently as non-curious agents with increasing preference for the reward.

5.3 Simulation 3: Long-Term Success

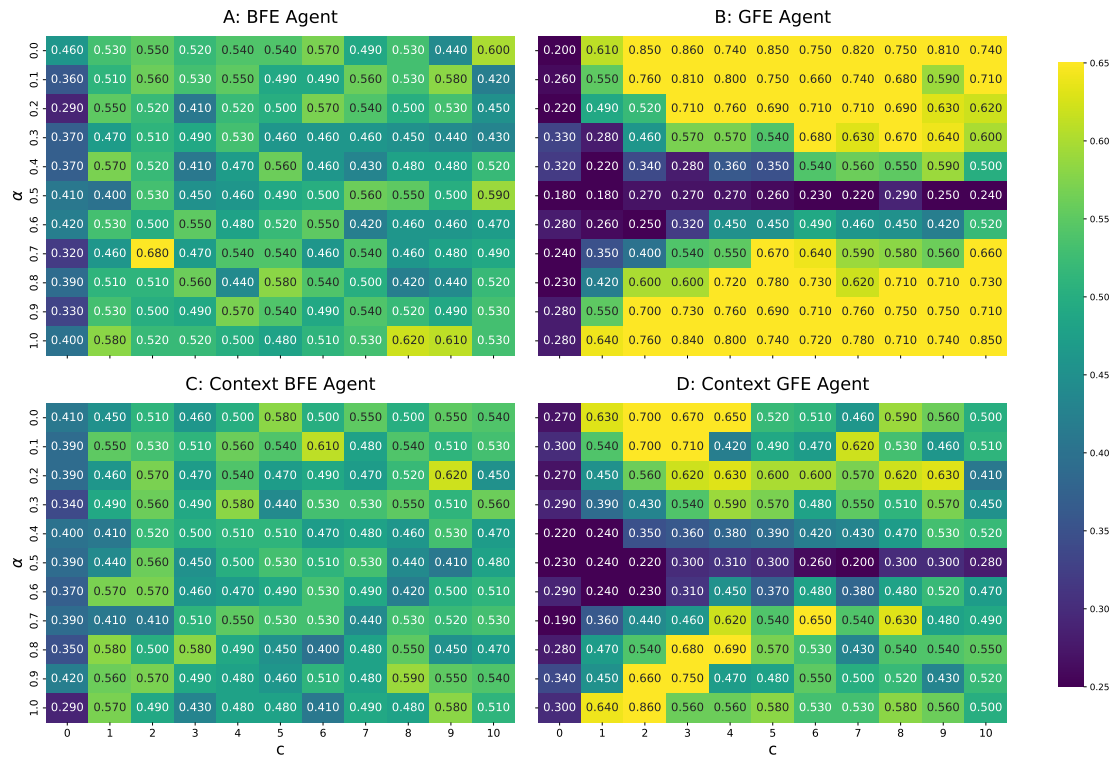


Figure 5.3: Ratios of receiving the reward in timestep $t = 1$ or $t = 2$ over 1000 trials for the four agents (BFE, GFE, Context-BFE, and Context-GFE) with differing α - and c values.

The values in the heatmaps' cells give the ratios of receiving the reward in timestep $t = 1$ or $t = 2$ over 1000 trials. Again, the four plots reveal an axial symmetry around $\alpha = 0.5$. That means both alpha values (α and $1 - \alpha$) lead to similar ratios of receiving the reward for all four agents.

Moreover, we can observe the behavioral divide between non-curious agents and curious agents also in this final experiment. Because the presented data does not represent agentic beliefs but empirical outcomes of 1000 trials, both agent types yield

qualitatively similar but not equal results. The differences between non-curious and curious agents are given in the following.

Overall, non-curious agents have higher rates of receiving the reward than curious agents for almost all combinations of α - and c values except for large c values that are also paired with extreme α values (e.g., $p = 0.607$ at $(\alpha = 0.0, c = 10)$ in Fig. 5.3B vs. $p = 0.558$ at $(\alpha = 0.0, c = 10)$ in Fig. 5.3D)), i.e., curious agents only achieve to receive the reward more frequently than non-curious agents when they have a strong preference for observing the reward and also knowing that the T-Maze type (reward in left arm or reward in right arm) is strongly correlated with actually observing the reward at the assumed location.

For all four agents it is a general trend that larger c values together with more extreme α values lead to similar or higher ratios of receiving the reward.

6 Discussion

In this chapter I interpret the findings of the four agents and then argue that the newly developed Context-GFE agent shows to kind of behavior that is expected from a curious agent that seeks information in its environment. Point of reference for this behavior is the GFE agent. In the following non-curious agents refer to the BFE- and Context-BFE agents, while curious agents refer to the GFE- and Context-GFE agents.

6.1 Simulation 1: Initial Curiosity

In this section I analyze the results of Fig. 5.1. Here, the behavioral modes of all agents can be divided along two dimensions (not to be confused with the two axes in each plot): First, all agents are driven by a (varying) desire for observing the reward. Second, the curious agents also "want" to reduce uncertainty about hidden states (including context states) of their generative models. Both behavioral drives are investigated in the following.

6.1.1 Desire for observing the Reward

The agents' preference for observing the reward is directly encoded by the parameter c of their generative models. $c = 0$ means the agent does not prefer the reward over other observations, i.e., it expects to receive all possible observations equally often. Conversely, $c = 10$ means the agent has a strong preference for observing the reward and thus expects to receive it at (almost) every observation. All agents know the reward is located in one of the two arms in the T-Maze but certainly not at the two remaining locations (start and cue). When the agent has a stronger preference for the reward (higher c value) then it is also more likely to directly move to one of two arms and effectively gambles for the reward (there is a 50 % chance of observing the reward). In return the probabilities of performing an alternative action (e.g., "move to cue", or "move (stay) at start") become smaller. This consequence is evident in each agent type, where the probabilities of moving to the cue become smaller for larger c values.

6.1.2 Desire for reducing Uncertainty

The drive for reducing uncertainty about hidden states (including context states) is only evident in the curious agents. In the following, I explain the curiosity mechanism and how it relates to the results of the agents. High uncertainty levels of hidden states in

the generative model relates to the agent's assumption that some hidden causes of observations in the environment are not clear, meaning the agent assigns similar (highly entropic) probabilities to the different possible causes. In this simulation, the hidden causes of observations consist of the four T-Maze fields (left arm, right arm, start, and cue) and the two T-Maze types (type A, and type B). Regarding the fields, they are immediately disclosed at each observation, e.g., when the agent is at the start it immediately knows (through the observation) that it is at the start field. In the case of the T-Maze types observations do not regularly reveal the type of the maze. This situation is mirrored in the agents' belief distributions about hidden states where the states concerning the T-Maze type initially are maximally uncertain, i.e., both possibilities (type A and type B) have equal (highly entropic) probabilities $p = 0.5$. The curiosity mechanism then aims at reducing this uncertainty (entropy). This is achieved by selecting actions that are expected to lead to observations that are informative about this hidden state. This requires further clarification: Observations can only be informative about hidden states when the agent has a model of the relation between that hidden state and possible observations, i.e, this relation must not be random. This circumstance can directly be exploited by the curious agent by moving to the cue location. There it receives an observation for which the agent possesses a deterministic model how this observation relates to the T-Maze type (one possible observation leads to $p = 1.0$ for type A and the other possible observation leads to $p = 1.0$ for type B). Besides directly moving to the cue location, there is a second option. The agent could also directly move to one of the two arms and thus observes the reward or no reward. This observation is also related to the T-Maze type but not deterministically but through the parameter α . This second option is only useful when $\alpha \stackrel{!}{=} 0.5$, i.e, when there is a non-random relation between the observation and the T-Maze type. Moreover, more deterministic values like $\alpha = 0.0$ or $\alpha = 1.0$ makes this observation (reward) more useful, where it can then infer that the maze has type A in the former case and type B for the latter. In this situation, receiving no reward in an arm is also informative because the agent can then infer it is interacting with the opposite maze type. This circumstance becomes evident in the results of both curious agents. Both agents are driven by their curiosity and can only satisfy this "desire" by either going to the cue or moving to one of the arms—but only in the case when it knows what an observation at an arm means in relation to the hidden T-Maze type, i.e, when $\alpha \stackrel{!}{=} 0.5$. In the most ambiguous case $\alpha = 0.5$, moving to an arm cannot satisfy the curiosity drive, i.e, it does not reduce uncertainty about the T-Maze type and thus the agent is then increasingly driven to reduce this uncertainty using the other option it still has: It instead moves to the cue. This trend is visible in the results of both curious agents where they initially move to the cue with highest probabilities ($p = 0.380$) for α values near 0.5.

6.1.3 Compromises of the Mean-Field Approximation in the Context-GFE agent

While the results of both curious agents are qualitatively similar, they are not equal, meaning that although the Context-GFE agent also performs information seeking like the GFE agent, there are subtle differences that are addressed in the following. The Context-GFE agent shows a more abrupt transition of probabilities from smaller c values and α values near 0.5 to larger c values and more extreme α values. This phenomenon is most likely a consequence of a mean-field approximation that was necessary to develop the Context-GFE agent. Specifically, while the GFE agent models the T-Maze type as part of its hidden state, the Context-GFE agent instead models it via its context state. The differences between the two approaches are as follows. The hidden state is modeled individually for each timestep in the trial (simulation) while there is only a single context state that persists for the whole trial (see ...). Furthermore, all three observations are related to this single context state in the agent's generative model, i.e., in the factor graph representation there are edges between all observation states and the context state. This leads to cycles in the factor graph which can lead to inference algorithms that do not converge to a stable solution in the case when the statistical dependencies between variables at each node are respected by the local belief distributions, i.e., when we assume the Bethe free energy at each local node. As a remedy, it is sufficient to break these local statistical dependencies at single nodes to remove the cycles. This was done between the observation states and the context state and leads to mean-field approximations at these local nodes. As a result, the inference process leads to stable solutions but has the problem of disrespecting uncertainties in the generative model related to context- and observation states. As a consequence, inference can yield results that are overconfident and thus are not as nuanced as without using the mean-field approximation. This circumstance becomes evident in the results of both curious agents where the GFE agent does not rely on the mean-field approximation between its observation states and hidden states (it does not have a context state and instead models the T-Maze type via the hidden states) and thus achieves a nuanced behavioral shift with varying α - and c values. In contrast, the Context-GFE agent relies on the mean-field approximation at the prescribed locations in its generative model and becomes overly confident more quickly with varying parameter combinations (α and c) which leads to the visible abrupt change of probabilities for larger c values and extreme α values.

6.2 Simulation 2: Exploiting Information

In this section I interpret the results of Fig. 5.2. Again, the following behavioral analysis can be divided into two parts that first relate to all agents' drive to receive the reward, and second relate to the curious agents' drive to also seek information in the environment.

6.2.1 Moving to the cued Reward Field

All four agents show similar characteristics of moving to the arm field that is suggested by the cue. Specifically, the cue reveals the type of the T-Maze (type A or type B) and the agents can relate this information to the specific arm that is expected to contain the reward via the parameter α . Extreme α values allow for a more certain prediction of the cue-containing arm field. This situation is evident in all four agents. The probabilities of moving to the arm that is suggested by the cue increase with more certain (extreme) α values, i.e., the agents know there is a strong correlation between the T-Maze type and the location of the reward (left arm or right arm) and exploit this knowledge to find the reward.

Similarly, the c values describe the agents' preferences of observing the reward, where higher values lead to stronger preferences. This circumstance is visible in the results of all four agents where they all are more likely to move to the cued reward location with increasing c values, i.e., the more they "want" to receive the reward the more likely they exploit the information from the cue.

6.2.2 Curious Agents still seek Information

Both curious agents have similar results compared to the non-curious agents with subtle differences that seem to relate to their intrinsic drive to continue seeking information in the environment (even after receiving the cue). The curious agents yield lower probabilities of moving to the cued location for most α - and c values, i.e., in these cases the curious agents assign more probability mass to other actions which indicates that these still contain information that the agents seek. The few configurations of α - and c values that have higher probabilities than the non-curious agents hint at the opposite: There the curious agents seem to assume information at these cued fields that can be exploited. These findings require further investigation.

6.3 Simulation 3: Long-Term Success

In this section I interpret the results of Fig. 5.3. As for previous simulations the results show a behavioral divide between non-curious and curious agents. These are detailed in the following.

6.3.1 Success Factors

First, I consider the trends that are evident in all four agents. Generally, higher c values and more extreme α values lead to higher success rates of receiving the reward in a trial. This is plausible since higher c values increase the agents' drive of pursuing the reward and more extreme α values render the cue more useful for moving to the correct arm that contains the reward.

6.3.2 An Alternative Meaning of Success

The curious agents appear to have lower success rates for most combinations of α - and c values. In these cases their drive of curiosity seem to lead them to alternative behavioral pathways such that they miss out on the reward more regularly. This situation should not be mistaken to be a weakness of curious agents, instead it shows their inclination of not only preferring predicted observations (like non-curious agents do) but also value hidden environmental causes that are less ambiguous, i.e, they seek to reduce uncertainty about their model of these causes, which are the hidden states. This mechanism may allow them to reduce their variational free energy more reliably in the long term and thus better secure their existence in evolutionary terms (according to the Free Energy Principle). In this way, the term "Long-Term Success" might not only be understood as the rates of receiving the reward but additionally refers to the agent's ability to minimize its variational free energy which necessitates the curiosity mechanism.

6.4 Summary

In conclusion, the newly developed Context-GFE agent shows analog behavior to the GFE-Agent that serves as a reference for a curious agent. Importantly, both curious agents exhibit significantly different behaviors compared to non-curious agents, which can be attributed to their drive of seeking information in the environment.

6.5 Related Work

7 Conclusion